

Universität Stuttgart

Institut für Ingenieurgeodäsie



Integration of Map Data into a Constrained Kalman Filter as an Additional Sensor

Master's Thesis

Ekrem Serdar Ozturk

Registration Number: 3643781

Geomatics Engineering (GEOENGINE)

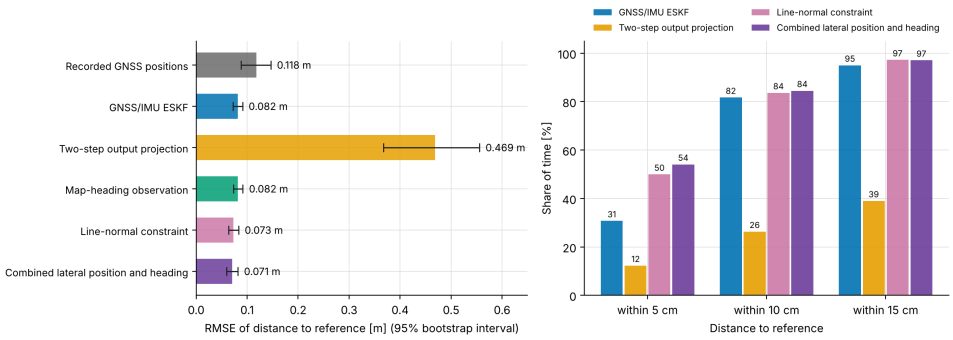
University of Stuttgart
Faculty of Aerospace Engineering and Geodesy
Institute of Engineering Geodesy

Examiner:	Dr.-Ing. Li Zhang
Supervisor:	M.Sc. Rudolf Frolow

Date of Submission: 11 November 2026

Management Summary

This thesis treats map data as an additional sensor in a GNSS/IMU error-state Kalman filter (ESKF): instead of snapping the finished trajectory to a map in a separate step, the map enters the filter as an observation. Two maps are used, OpenStreetMap (OSM) and a track polyline digitized from true orthophotos. Each provides three observations: the track heading, the lateral (line-normal) distance to the track, and both combined. They are compared with the GNSS/IMU filter without map and with a conventional two-step projection on a railway recording of about 9 km with RTK GNSS and a 300 Hz IMU.



Continuous GNSS: RMSE of the distance to the evaluation reference with 95% bootstrap intervals (left) and share of time within 5, 10 and 15 cm of it (right).

With continuous RTK GNSS, the filter reports the point on the track 3 m below the roof antenna. This alone lowers the error against the digitized evaluation reference from 0.118 m for the recorded antenna positions to 0.082 m, because the antenna swings sideways when the vehicle rolls on canted track. The OSM line-normal observation lowers it further to 0.073 m and the combined observation to 0.071 m (13 %), by up to 41 % in the curves with the largest errors. The heading observation does not move the position but lowers the heading error from 1.89 ° to 0.49 ° (74 %). Snapping the output to OSM in two steps raises the error to 0.469 m, because the result inherits the error of the map.

In five 60 s GNSS outages the filter without map runs on the IMU alone and drifts sideways by 35 m RMSE. The line-normal observation keeps this cross-track error at 0.55 m with OSM and 0.57 m with the orthophoto map, about 98 % less and below 1 m in every window, on straight track and in curves. Where the orthophoto map is complete it reaches 0.13 m to 0.14 m. A track line does not tell the filter where along the track it is, so the along-track error stays large, and the combined observations lose the map in the sharp curve of one window. Integrating the map in the filter is therefore most useful when GNSS is degraded or missing.

Keywords: GNSS/IMU integration, error-state Kalman filter, constrained estimation, map matching, OpenStreetMap, orthophoto, GNSS outage, railway localization

Universität Stuttgart

Institut für
Ingenieurgeodäsie

Geschwister-Scholl-Str. 24/D
70174 Stuttgart
Tel. 0711 / 685 - 84041
Fax 0711 / 685 - 84044
E-Mail: sekretariat@iigs.uni-stuttgart.de
Web: www.uni-stuttgart.de/IIGS/

MASTERTHESIS

for Ekrem Serdar Ozturk, Matr.-No. 3643781

Title: Integration of map data into a constrained Kalman Filter as an additional sensor

Sensor fusion algorithms are commonly used to combine data from a variety of sensors. In the context of kinematic positioning, the most common combination is GNSS and IMU. Map-matching is used to integrate the map data to improve the trajectories in two steps. These two steps represent the current state of the art. This thesis focuses on integrating map data directly into the sensor fusion process to implement a one-step approach. Therefore, the map will be considered an additional sensor providing data for the filter algorithm. First, investigations into existing sensor fusion algorithms with map integration must be performed. Next, different possibilities for constraints (e.g., based on coordinates, orientation) to be added to the sensor fusion algorithms will be researched. These constraints will then be implemented with help of map data provided by OpenStreetMap and orthophotos, and their effect on the solution evaluated. The resulting trajectories will be evaluated against given reference data.

Working Steps:

- Literature research on sensor fusion algorithms with map integration
- Implementation of a one-step approach with different constraints (e.g., based on coordinates, orientation) from map data (OpenStreetMap, Orthophotos)
- Evaluation of the implemented algorithms with given reference data

Deliverables:

- Master thesis in analogue and digital format
- Implemented algorithms

Supervisor: M. Sc. Rudolf Frolow
Examiner: Dr.-Ing. Li Zhang

Start Date: 11.05.2026
Deadline: 11.11.2026

(Dr.-Ing. Li Zhang)

Declaration of Originality

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1 Introduction

1.1 Motivation

Reliable pose estimation is required by autonomous vehicles, mobile mapping systems, and rail-bound platforms. An inertial navigation system propagates position, velocity, and orientation at high rate, but small sensor biases are integrated over time and cause growing navigation errors. GNSS measurements limit this drift when reception is good, although their rate is lower than the IMU rate and their quality can deteriorate because of blockage, multipath, or outliers. These complementary properties motivate GNSS/IMU fusion (Groves 2013; Titterton and Weston 2004).

A digital map provides information that a conventional GNSS/IMU filter does not use. If a platform travels on a road or railway, its position should remain near the mapped centre line and its direction should normally agree with the local route direction. These relations are particularly informative laterally: GNSS errors and inertial drift can move an estimate across neighbouring tracks even though the real platform remains constrained to its route.

Map information may be used after estimation or inside the estimator. A two-step method projects a completed trajectory onto the map but does not correct the internal state or covariance. A one-step method treats the map relation as a measurement. The correction can then affect position, orientation, velocity, and sensor-bias errors through their cross-covariances, and the updated uncertainty is propagated to the next epoch. This feedback is the main motivation for the present work.

The thesis considers map information from OpenStreetMap (OSM) and georeferenced orthophotos, as required by the thesis assignment. Existing OSM nodes and ways in the map database are reconstructed as ordered polylines from their stored UTM coordinates. The orthophoto source is a track polyline digitized from true orthophotos, which is processed into a second map in the same metric frame. The central question is how coordinate and orientation information from these map sources can be represented as statistically weighted observations and integrated directly into the filter.

1.2 Problem Statement

This thesis investigates whether map information improves a GNSS/IMU pose estimate when it is introduced as a statistically weighted constraint in an error-state Kalman filter (ESKF). In this thesis, ESKF denotes an extended Kalman filter implemented with a nonlinear nominal state and a small error state. Thus, an ESKF belongs to the EKF family; the additional term “error-state” specifies the state representation and correction procedure. It does not denote an iterated Kalman filter. The implemented 18-state quaternion ESKF was developed for this thesis. The task is not only to select the nearest segment. A valid update must identify a plausible

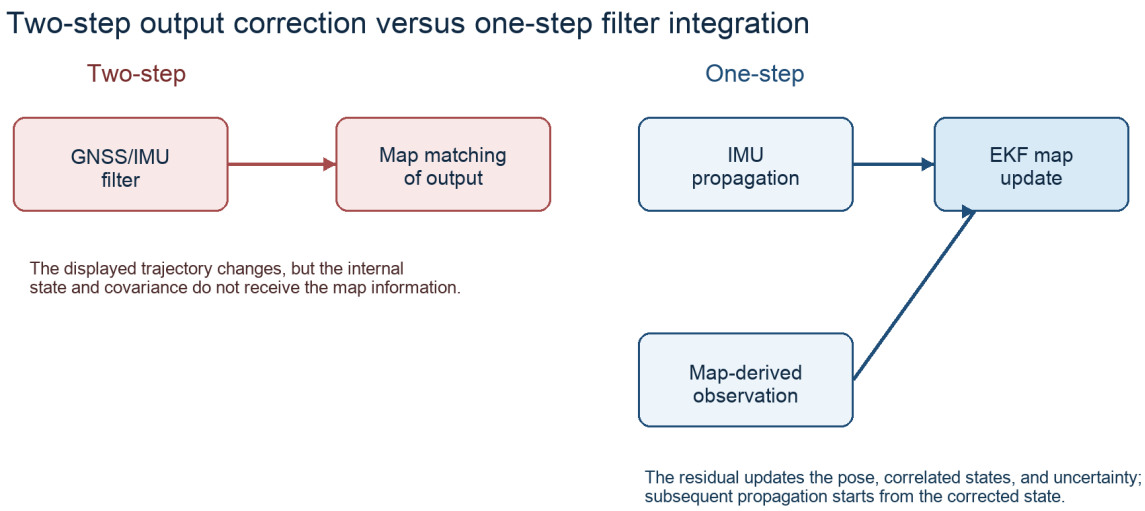


Figure 1.1 Conceptual difference between output-only map matching and direct map-aided filtering.

map segment, define a residual and Jacobian, assign realistic uncertainty, reject inconsistent or ambiguous associations, and preserve filter consistency.

The recorded track dataset represents fast motion along a predominantly track-constrained route. It exposes important failure cases such as parallel tracks, switches, imperfect map geometry, and predicted positions that are far enough from the true route for the association procedure to select a neighbouring or otherwise incorrect segment. The study compares three representations of map information, namely a coordinate-based constraint, an orientation constraint, and their combination. Their corrections are evaluated relative to GNSS/IMU fusion and output-only map matching. The sensor input, initialization, evaluation interval, and applicable association procedure are held constant so that the comparison isolates the effect of the constraint formulation and map source.

1.3 Research Objectives

The objectives are to:

1. formulate a reproducible GNSS/IMU ESKF with explicit propagation, covariance, and correction models;
2. derive map information from OSM and orthophotos in a coordinate frame compatible with the navigation state;
3. reconstruct route geometry from OSM nodes and ways and associate the predicted pose with a locally plausible segment;
4. implement and compare heading, lateral-position, and combined lateral-position and heading constraints with uncertainty models and innovation gates;
5. compare one-step integration, which corrects the filter state and covariance, with two-step map matching, which changes only the reported trajectory;
6. evaluate the applicable constraint configurations against the same reference data under identical GNSS/IMU inputs; and

7. report accuracy, robustness, update acceptance, and consistency without using the evaluation reference as a filter measurement.

1.4 Contributions

The first contribution is the formulation and implementation of three map-derived observations within a GNSS/IMU ESKF. The heading observation corrects yaw without forcing position onto the centre line. The lateral observation constrains cross-track position. The combined observation stacks the lateral-position and heading residuals in one filter update.

The second contribution is a controlled comparison of these observations. All configurations use the same IMU propagation, GNSS model, initialization, coordinate frame, map-association procedure, and evaluation interval. The comparison therefore tests the effect of the information supplied by each constraint under controlled conditions.

The third contribution is a reproducible procedure for reconstructing the OSM ways, rejecting distant or ambiguous associations, and evaluating the accepted corrections against independent reference data. The same constraints are applied to an orthophoto-derived track map. The OSM constraints are tested under continuous GNSS against a digitized evaluation reference, and both map sources are compared during predefined GNSS outages, where the withheld GNSS positions serve as an independent truth.

1.5 Thesis Structure

Chapter 2 introduces inertial navigation, sensor fusion, Kalman filtering, extended Kalman filtering, and constrained state estimation. Chapter 3 reviews map matching, map-aided localization, road-constrained localization, and direct map integration in Kalman filters. Chapter 4 defines the system architecture, state model, map representation, constraints, and proposed one-step update. Chapter 5 documents the dataset, OSM and orthophoto processing, constraint generation, filter implementation, and software architecture. Chapter 6 specifies the test environment, reference data, evaluation metrics, and baseline approaches. Chapter 7 compares the trajectories, position and orientation accuracy, and the effects of the position and orientation constraints, and Chapter 8 summarizes the findings, limitations, and future work.

2 Theoretical Background

2.1 Inertial Navigation Systems

A strapdown inertial navigation system numerically integrates the angular rate measured by gyroscopes and the specific force measured by accelerometers. The sensors are fixed to the platform body frame b; their measurements are rotated into a world frame w by the estimated attitude. Standard derivations can be found in Groves (2013) and Titterton and Weston (2004). This thesis uses a quaternion for the nominal attitude because it avoids the singularities of a three-angle parameterization during propagation, while small attitude errors are represented by a three-dimensional rotation vector in the error state (Solà 2017).

To avoid ambiguity, a subscript “meas” is attached to the sensor output and “true” to the ideal physical quantity. With additive bias and white measurement noise, the gyroscope and accelerometer models are

$$\boldsymbol{\omega}^{\mathrm{b}}_{ib,\mathrm{meas}} = \boldsymbol{\omega}^{\mathrm{b}}_{ib,\mathrm{true}} + \boldsymbol{b}_g + \boldsymbol{n}_g, \tag{2.1}$$

$$\boldsymbol{f}^{\mathrm{b}}_{ib,\mathrm{meas}} = \boldsymbol{f}^{\mathrm{b}}_{ib,\mathrm{true}} + \boldsymbol{b}_a + \boldsymbol{n}_a. \tag{2.2}$$

Here, the subscript ib means that the body frame b is described with respect to the inertial frame i, while the superscript b means that the quantity is expressed in body-frame coordinates. Thus, $\boldsymbol{\omega}^{\mathrm{b}}_{ib}$ is the angular velocity of the body frame with respect to the inertial frame, expressed in b, and $\boldsymbol{f}^{\mathrm{b}}_{ib}$ is the corresponding specific force expressed in b. The latter is force per unit mass excluding gravity, which is the physical quantity measured by an accelerometer. It is therefore denoted by $\boldsymbol{f}$, not by $\boldsymbol{a}$. The world-frame kinematic acceleration is denoted by $\boldsymbol{a}^{\mathrm{w}}$ after gravity has been added in equation (2.4). The vectors $\boldsymbol{b}_g$ and $\boldsymbol{b}_a$ are the gyroscope and accelerometer biases; and $\boldsymbol{n}_g$ and $\boldsymbol{n}_a$ are zero-mean measurement noise. A tilde is frequently used for measured quantities in the literature. The explicit “meas” label is retained because it remains readable when equations are copied between software and manuscript notation.

The bias-corrected measurements are

$$\widehat{\boldsymbol{\omega}}^{\mathrm{b}} = \boldsymbol{\omega}^{\mathrm{b}}_{ib,\mathrm{meas}} - \widehat{\boldsymbol{b}}_g, \qquad \widehat{\boldsymbol{f}}^{\mathrm{b}} = \boldsymbol{f}^{\mathrm{b}}_{ib,\mathrm{meas}} - \widehat{\boldsymbol{b}}_a, \tag{2.3}$$

and the world-frame acceleration becomes

$$\boldsymbol{a}^{\mathrm{w}} = \boldsymbol{R}^{\mathrm{w}}_{\mathrm{b}}(\widehat{\boldsymbol{q}}_{wb})\widehat{\boldsymbol{f}}^{\mathrm{b}} + \boldsymbol{g}^{\mathrm{w}}. \tag{2.4}$$

The corrected measurements in equation (2.3) drive the nominal state. The implementation combines consecutive IMU samples by midpoint integration. For a time interval Δt, velocity and position are updated using the estimated acceleration, while the quaternion is updated by the

exponential map of the corrected angular increment. Midpoint integration reduces the first-order timing error that would arise from using only the beginning of the interval.

Biases are not assumed to be perfectly known. Their residual evolution is modelled as a random walk, which allows the filter to adjust them when external measurements create observable correlations. Without such measurements, position and orientation errors grow because bias errors are repeatedly integrated. This is the fundamental reason that low-rate absolute or map-relative information can be valuable even when the IMU rate is high.

Table 2.1 Principal navigation symbols and frames.

Symbol	Meaning	Unit
$\boldsymbol{q}_{wb}$	Quaternion rotating vectors from body frame b to world frame w	–
$\boldsymbol{p}^{\mathrm{w}}$	Platform position in the world/UTM frame	m
$\boldsymbol{v}^{\mathrm{w}}$	Platform velocity in the world frame	m/s
$\boldsymbol{b}_g$	Gyroscope bias	rad/s
$\boldsymbol{b}_a$	Accelerometer bias	m/s^2
$\boldsymbol{g}^{\mathrm{w}}$	Gravity represented in the world frame	m/s^2
$\boldsymbol{R}_{\mathrm{b}}^{\mathrm{w}}$	Body-to-world rotation matrix	–

2.2 Sensor Fusion

Sensor fusion combines complementary information while accounting for uncertainty. The IMU supplies high-rate relative motion but no absolute position. GNSS supplies global position at a lower rate and with environment-dependent errors. A map supplies geometric or appearance-based relations whose precision depends on map quality and association validity. A probabilistic estimator should therefore not treat any of these sources as exact.

The loose and tight integration distinction is described by Groves (2013) and evaluated experimentally by Falco et al. (2017). A loosely coupled GNSS/IMU system uses a position or velocity solution already computed by the GNSS receiver. A tightly coupled system instead processes individual satellite observables, for example pseudoranges and Doppler or carrier-phase measurements, together with the inertial data. The terms “loose” and “tight” describe the interface between the GNSS receiver and the navigation filter; they do not describe how strongly a map constraint is weighted. The dataset used in this thesis provides GNSS positions rather than individual satellite observables. The implemented GNSS/IMU integration is therefore loosely coupled. Map-derived quantities are added as further filter observations. They are often called pseudo-measurements because they are derived from prior map information and the current association, not because they are arbitrary or fabricated.

Timing and coordinate consistency are as important as the estimator equations. IMU samples must be propagated in chronological order, and the state must be evaluated at the observation time before an update is applied. Geographic GNSS coordinates and mapped geometry are converted to a common metric UTM frame. For the track dataset, IMU data are interpolated to exact GNSS timestamps when required by the data interface. This interpolation does not create an independent sensor observation. It approximates the angular rate and specific force between the two surrounding IMU samples so that the final partial propagation interval ends at the GNSS timestamp. All original IMU samples still provide the high-rate propagation. Applying

a GNSS update directly at the nearest IMU epoch would instead introduce a time offset unless both timestamps already coincide.

2.3 Kalman Filtering

For a linear system with Gaussian process and observation noise, the Kalman filter recursively computes the minimum-variance linear estimate (Kalman 1960). Let $\boldsymbol{x}_k$ be the state at epoch k, that is, at the k-th time at which a measurement is processed, $\boldsymbol{F}_k$ the transition matrix, $\boldsymbol{u}_k$ a known input, and $\boldsymbol{z}_k$ a measurement. The discrete process and observation models are

$$\boldsymbol{x}_k = \boldsymbol{F}_k \boldsymbol{x}_{k-1} + \boldsymbol{B}_k \boldsymbol{u}_k + \boldsymbol{w}_k, \qquad \boldsymbol{w}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_k), \tag{2.5}$$

$$\boldsymbol{z}_k = \boldsymbol{H}_k \boldsymbol{x}_k + \boldsymbol{v}_k, \qquad \boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_k). \tag{2.6}$$

The prediction equations are

$$\widehat{\boldsymbol{x}}_k^- = \boldsymbol{F}_k \widehat{\boldsymbol{x}}_{k-1}^+ + \boldsymbol{B}_k \boldsymbol{u}_k, \tag{2.7}$$

$$\boldsymbol{P}_k^- = \boldsymbol{F}_k \boldsymbol{P}_{k-1}^+ \boldsymbol{F}_k^\mathsf{T} + \boldsymbol{Q}_k. \tag{2.8}$$

With innovation $\boldsymbol{r}_k$ and innovation covariance $\boldsymbol{S}_k$, the update is

$$\boldsymbol{r}_k = \boldsymbol{z}_k - \boldsymbol{H}_k \widehat{\boldsymbol{x}}_k^-, \qquad \boldsymbol{S}_k = \boldsymbol{H}_k \boldsymbol{P}_k^- \boldsymbol{H}_k^\mathsf{T} + \boldsymbol{R}_k, \tag{2.9}$$

$$\boldsymbol{K}_k = \boldsymbol{P}_k^- \boldsymbol{H}_k^\mathsf{T} \boldsymbol{S}_k^{-1}, \qquad \widehat{\boldsymbol{x}}_k^+ = \widehat{\boldsymbol{x}}_k^- + \boldsymbol{K}_k \boldsymbol{r}_k. \tag{2.10}$$

The covariance is updated in Joseph form,

$$\boldsymbol{P}_k^+ = (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k) \boldsymbol{P}_k^- (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k)^\mathsf{T} + \boldsymbol{K}_k \boldsymbol{R}_k \boldsymbol{K}_k^\mathsf{T}. \tag{2.11}$$

The Joseph expression is algebraically equivalent to the shorter covariance update under exact arithmetic, but it better preserves symmetry and positive semidefiniteness in finite precision (Bar-Shalom et al. 2001; Brown and Hwang 2012). The implementation uses this form for GNSS and map updates.

2.4 Error-State Kalman Filter

Navigation dynamics and map observations are nonlinear. The extended Kalman filter replaces the linear transition and observation matrices with Jacobians of nonlinear functions about the current estimate. Directly filtering a quaternion is possible, but the unit-norm constraint complicates an additive correction. An error-state formulation instead propagates a nonlinear nominal state and applies a small correction in a locally Euclidean error space (Solà 2017).

This thesis calls the resulting estimator an error-state Kalman filter (ESKF). Its nonlinear process and observation models are linearized in the same manner as an extended Kalman filter. No repeated relinearization is performed within a measurement update, so the estimator is not an iterated Kalman filter.

The active navigation state used in this thesis contains nominal attitude, position, velocity, gyroscope bias, accelerometer bias, and gravity. Its small error vector is

$$\delta \boldsymbol{x} = \begin{bmatrix} \delta \boldsymbol{\theta}^\mathsf{T} & \delta \boldsymbol{p}^\mathsf{T} & \delta \boldsymbol{v}^\mathsf{T} & \delta \boldsymbol{b}_g^\mathsf{T} & \delta \boldsymbol{b}_a^\mathsf{T} & \delta \boldsymbol{g}^\mathsf{T} \end{bmatrix}^\mathsf{T}. \tag{2.12}$$

The continuous error dynamics and covariance dynamics are

$$\delta \dot{\boldsymbol{x}} = \boldsymbol{F}_c \delta \boldsymbol{x} + \boldsymbol{G}_c \boldsymbol{n}, \tag{2.13}$$

$$\dot{\boldsymbol{P}} = \boldsymbol{F}_c \boldsymbol{P} + \boldsymbol{P} \boldsymbol{F}_c^\mathsf{T} + \boldsymbol{G}_c \boldsymbol{Q}_c \boldsymbol{G}_c^\mathsf{T}. \tag{2.14}$$

The implementation uses the first-order discretization

$$\boldsymbol{F}_k \approx \boldsymbol{I} + \boldsymbol{F}_c \Delta t, \qquad \boldsymbol{Q}_k \approx \boldsymbol{G}_c \boldsymbol{Q}_c \boldsymbol{G}_c^\mathsf{T} \Delta t. \tag{2.15}$$

After an accepted measurement update, position, velocity, biases, and gravity receive additive corrections. The attitude error is injected multiplicatively through a small rotation quaternion. The error mean is then reset to zero and the covariance is reset consistently with the changed error coordinates (Solà 2017).

Linearization is only locally valid. A map association that is geometrically wrong can therefore be more damaging than an ordinary noisy measurement: the residual may be computed with respect to the wrong road or track. Association checks and innovation gating are consequently part of the observation model, not optional post-processing.

2.5 Constrained State Estimation

State constraints can be enforced by projection, by augmenting the measurement model, or by reparameterizing the state. Equality-constrained Kalman filters and their nonlinear extensions show how known relations can reduce uncertainty when handled consistently (Gupta and Hauser 2007; Julier and LaViola 2007; Simon and Chia 2002). Map geometry is not exact enough to be imposed as a hard equality in this application. It is therefore modelled as a soft measurement with a covariance that represents map accuracy, platform width, association uncertainty, and unmodelled behaviour.

A nonlinear map relation $\boldsymbol{h}_m(\boldsymbol{x})$ and map-derived observation $\boldsymbol{z}_m$ yield

$$\boldsymbol{r}_m = \boldsymbol{z}_m - \boldsymbol{h}_m(\widehat{\boldsymbol{x}}^-), \tag{2.16}$$

$$\boldsymbol{H}_m = \left. \frac{\partial \boldsymbol{h}_m}{\partial \delta \boldsymbol{x}} \right|_{\delta \boldsymbol{x} = \boldsymbol{0}}. \tag{2.17}$$

The innovation covariance and normalized innovation squared (NIS) are

$$\boldsymbol{S}_m = \boldsymbol{H}_m \boldsymbol{P} \boldsymbol{H}_m^{\mathsf{T}} + \boldsymbol{R}_m, \qquad \mathrm{NIS} = \boldsymbol{r}_m^{\mathsf{T}} \boldsymbol{S}_m^{-1} \boldsymbol{r}_m. \tag{2.18}$$

If the modelling assumptions are adequate, the NIS follows a chi-square distribution with degrees of freedom equal to the residual dimension (Bar-Shalom et al. 2001). The implemented one-dimensional heading and lateral updates use the 99% threshold 6.6349; the two-dimensional combined updates use 9.2103. These gates reject observations inconsistent with the predicted state and uncertainty. They do not prove that an accepted association is correct, so geometric ambiguity is checked separately.

3 State of the Art

This chapter positions the thesis within four related areas: conventional map matching, map-aided localization, road- and track-constrained estimation, and direct map integration in recursive filters. The distinction between these areas is important. A method that only assigns an estimated point to a road is not equivalent to a method that feeds map information back into the navigation state and its covariance.

3.1 Map Matching

Map matching determines which element of a transport network corresponds to a measured or estimated position. The input may be an isolated GNSS fix, a sequence of positions, or the output of a navigation filter. The map usually contains nodes, polylines, and topological connections. The result may be a matched road or track identifier, a curvilinear coordinate along that element, or a corrected Cartesian position.

Quddus, Ochieng, and Noland (2007) divide map-matching methods into geometric, topological, probabilistic, and advanced approaches. Geometric methods use quantities such as point-to-point or point-to-curve distance. Topological methods additionally use connectivity and the continuity of the route. Probabilistic methods represent measurement uncertainty and rank alternative links by likelihood. The earlier algorithm of Quddus, Ochieng, Zhao, et al. (2003), for example, combines geometric and topological information rather than accepting the closest road unconditionally.

For a point $\boldsymbol{p}$ and line segment with endpoints $\boldsymbol{a}$ and $\boldsymbol{b}$, the standard geometric projection is

$$u = \frac{(\boldsymbol{p}-\boldsymbol{a})^{\mathsf{T}}(\boldsymbol{b}-\boldsymbol{a})}{\|\boldsymbol{b}-\boldsymbol{a}\|^2}, \tag{3.1}$$

$$\bar{u} = \min(1, \max(0, u)), \tag{3.2}$$

$$\boldsymbol{c} = \boldsymbol{a} + \bar{u}(\boldsymbol{b}-\boldsymbol{a}), \tag{3.3}$$

$$d = \|\boldsymbol{p}-\boldsymbol{c}\|. \tag{3.4}$$

Here, $\boldsymbol{c}$ is the closest point on the finite segment and d is the point-to-segment distance. Selecting the segment with minimum d is fast, but it uses only one epoch and can switch between neighbouring links whenever their distances are similar.

Heading agreement provides a second geometric cue. If ψ_i is the tangent heading of segment i and $\widehat{\psi}$ is the predicted platform heading, an undirected segment must be compared in both directions:

$$\Delta\psi_i = min\Big(\Big|\mathrm{wrap}(\psi_i - \widehat{\psi})\Big|, \Big|\mathrm{wrap}(\psi_i + \pi - \widehat{\psi})\Big|\Big). \tag{3.5}$$

The second term is necessary because the same geometric line can be traversed in the opposite direction. This ambiguity is especially relevant for railway ways, for which the ordering of stored nodes does not by itself prove the current direction of travel.

Sequence-based methods incorporate temporal continuity. In the hidden Markov model of Newson and Krumm (2009), candidate road positions are hidden states, the distance between a measurement and a candidate contributes an emission probability, and the consistency between successive candidates contributes a transition probability. A typical maximum-a-posteriori sequence can be written as

$$\widehat{\boldsymbol{s}}_{1:N} = \arg\max_{\boldsymbol{s}_{1:N}} p(\boldsymbol{s}_1) p(\boldsymbol{z}_1 \mid \boldsymbol{s}_1) \prod_{k=2}^{N} p(\boldsymbol{s}_k \mid \boldsymbol{s}_{k-1}) p(\boldsymbol{z}_k \mid \boldsymbol{s}_k). \tag{3.6}$$

Such models are more robust to sparse and noisy positions, although they add a discrete inference problem and may introduce a delay if a complete sequence is processed. The matcher in this thesis remains epoch-wise so that the effect of the observation model can be isolated. Its inability to maintain multiple route hypotheses is therefore an explicit limitation rather than an omitted assumption.

Conventional map matching often changes only the reported trajectory. If $\widehat{\boldsymbol{p}}_k$ is the filter output and $\boldsymbol{c}_k$ the closest map point, an output-only correction is simply

$$\widehat{\boldsymbol{p}}^{+}_{\mathrm{display},k} = \boldsymbol{c}_k, \qquad \widehat{\boldsymbol{x}}_k^{+} = \widehat{\boldsymbol{x}}_k^{-}, \qquad \boldsymbol{P}_k^{+} = \boldsymbol{P}_k^{-}. \tag{3.7}$$

The displayed path may follow the map, but the estimator continues from its uncorrected state. This architecture forms the two-step baseline used later.

3.2 Map-Aided Localization

Map-aided localization uses a map as prior information in the estimation process. Depending on the available map, the observable quantities may include road boundaries, lane markings, centre line position, heading, curvature, height, or topology. The present thesis uses only geometric information from already available OSM ways.

OSM represents points as nodes and linear features as ordered lists of nodes called ways (Haklay and Weber 2008). Its accessibility and broad coverage make it attractive for research and deployment. However, it is collaboratively maintained, and local geometry may originate from heterogeneous sources. Therefore, an OSM coordinate is not a deterministic truth measurement. The proper estimation model is

$$\boldsymbol{z}_m = h_m(\boldsymbol{x}, \mathcal{M}) + \boldsymbol{v}_m, \qquad \boldsymbol{v}_m \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_m), \tag{3.8}$$

where $\mathcal{M}$ denotes the map and $\boldsymbol{R}_m$ represents map geometry, association, and modelling uncertainty.

The uncertainty is directional. A route centre line gives strong information about lateral displacement if the vehicle is known to remain on that route, but an epoch-wise closest point gives much weaker information about progress along the route. In the local tangent-normal frame, a suitable covariance is

$$\boldsymbol{R}_{tn} = \begin{bmatrix} \sigma_{\parallel}^2 & 0 \\ 0 & \sigma_{\perp}^2 \end{bmatrix}, \qquad \sigma_{\parallel} > \sigma_{\perp}. \tag{3.9}$$

Rotating this covariance into the world frame retains the physical distinction between along-route and cross-route information. An isotropic position update would instead treat an arbitrary projection as if it were a complete planar position measurement.

Map quality and state uncertainty also influence data association. When the predicted covariance is large, several segments may be statistically plausible. A system that ignores this fact can become overconfident after choosing the wrong link. Robust designs therefore combine distance and heading tests, consider alternative candidates, and permit the matcher to abstain.

Orthophotos are a further map source. Unlike an OSM polyline, an orthophoto does not directly provide a vehicle-state observation. Observable content from an on-board sensor must first be related to the georeferenced aerial image. Vora et al. (2020) register LiDAR ground reflectivity against aerial imagery to estimate vehicle pose without requiring a separately surveyed high-definition reflectivity map. Multimodal similarity measures such as normalized mutual information are suitable when the two image sources do not have equal intensity statistics (Studholme et al. 1999; Veronese et al. 2015).

Learned cross-view approaches also treat orthophotos as global map information. Fervers et al. (2022) align on-board camera and LiDAR information with georegistered orthophotos for continuous metric localization. Their later uncertainty-aware vision-based method predicts a distribution over possible vehicle poses and integrates the predictions over time (Fervers et al. 2023). These methods demonstrate that an orthophoto constraint must include both a pose estimate and information about its reliability; a visually plausible registration maximum alone is not a sufficient filter observation.

Registration of on-board imagery against orthophotos requires a camera or LiDAR stream, which the railway recording does not contain. An orthophoto can, however, also be used offline: track geometry digitized from it provides a map with sub-metre accuracy in the same metric frame (Frolow et al. 2026). In this form, the orthophoto supplies the same coordinate and orientation information as the OSM polyline, but with a different accuracy and coverage. It can therefore use the same one-step ESKF correction mechanism, which permits a fair comparison of the two map sources while holding the GNSS/IMU estimator constant.

3.3 Road-Constrained Localization

Road-constrained estimation expresses known vehicle-motion or map relations as constraints on the state. Miller et al. (2007) treat the integration of GPS and digital road maps as a constrained-estimation problem. El Najjar and Bonnifait (2005) select the most likely road segment with belief theory and use it as a position observation in a Kalman filter. Its covariance is an ellipse aligned with the segment, small across the road and large along it; for a straight segment the along-road variance is infinite, so the map corrects only the transverse position. Fouque et al. (2008) use the direction of the matched navigable road as a heading observation in a Kalman filter and test the consistency of the map information before using it.

Constraints may be hard or soft. A hard equality has the form

$$c(\boldsymbol{x}) = 0, \tag{3.10}$$

and forces every accepted state to lie exactly on the constraint manifold. Projection methods and constrained Kalman filters can impose such equalities (Gupta and Hauser 2007; Julier and LaViola 2007; Simon and Chia 2002). A soft constraint instead assigns a nonzero covariance:

$$\mathbf{0} = c(\boldsymbol{x}) + \boldsymbol{v}_c, \qquad \boldsymbol{v}_c \sim \mathcal{N}(\mathbf{0}, \boldsymbol{R}_c). \tag{3.11}$$

Soft constraints are appropriate here because the platform can deviate from a mapped centre line and because OSM geometry and association are uncertain.

Railways make the value and difficulty of constrained localization especially clear. The motion is strongly restricted laterally, yet a railway network can contain parallel tracks and branching switches. Heirich (2016) model track identifier, along-track position, and train-to-track direction as different variables and use a particle representation to maintain competing track hypotheses. The work also reports that OSM-based track data should not be treated as highly accurate. Winter et al. (2019) evaluate train-borne GNSS/IMU localization using track-geometry constraints and report that the primary benefit is improved cross-track accuracy.

These findings lead to three consequences for this thesis. First, the route is not imposed as an exact one-dimensional state manifold. Second, the heading of an undirected segment is compared in both directions. Third, the single-Gaussian ESKF does not pretend to represent several track hypotheses: if competing ways have similar association scores, the map update is skipped.

3.4 Map Integration in Kalman Filters

In a recursive filter, a map relation can be handled as a pseudo-measurement. For predicted state $\widehat{\boldsymbol{x}}^-$ and covariance $\boldsymbol{P}^-$, the nonlinear map observation is linearized as

$$\boldsymbol{r}_m = \boldsymbol{z}_m - h_m(\widehat{\boldsymbol{x}}^-) \simeq \boldsymbol{H}_m \delta\boldsymbol{x} + \boldsymbol{v}_m. \tag{3.12}$$

The innovation covariance and Kalman gain are

$$\boldsymbol{S}_m = \boldsymbol{H}_m \boldsymbol{P}^- \boldsymbol{H}_m^\mathsf{T} + \boldsymbol{R}_m, \tag{3.13}$$

$$\boldsymbol{K}_m = \boldsymbol{P}^- \boldsymbol{H}_m^\mathsf{T} \boldsymbol{S}_m^{-1}. \tag{3.14}$$

The state correction $\boldsymbol{K}_m \boldsymbol{r}_m$ is distributed according to the covariance. Thus, even when a map residual directly observes only lateral position or yaw, correlated velocity and bias errors can also change.

The normalized innovation squared provides a consistency test:

$$\mathrm{NIS}_m = \boldsymbol{r}_m^\mathsf{T} \boldsymbol{S}_m^{-1} \boldsymbol{r}_m. \tag{3.15}$$

For an m-dimensional Gaussian innovation, NIS_m is compared with a chi-square threshold having m degrees of freedom. Rejection protects the filter from observations that are inconsistent with the modelled uncertainty, but it cannot prove that the chosen segment is correct. Data-association checks must therefore occur before innovation gating.

Direct integration also changes uncertainty. After a valid update, the posterior covariance is smaller in observed directions and is propagated by the next IMU step. Output snapping has no such effect. This difference is central when the experiment is intended to determine how each map constraint changes the estimated solution.

3.5 Existing One-Step Approaches

The literature uses several terms for direct map integration, including map-aided, map-constrained, tightly coupled, and road-centred estimation. In this thesis, *one-step* denotes an architecture in which the map residual updates the estimator state and covariance before the next propagation.

Fouque et al. (2008) use road-map direction as a heading observation and reject doubtful information through consistency tests. Tao et al. (2017) formulate a road-centred EKF that exploits road orientation and lane-related information while maintaining observability as the road frame changes. Atia and Waslander (2019) present adaptive map-aided GNSS/IMU fusion for urban navigation. In railway localization, Heirich (2016) demonstrate why multiple hypotheses are useful at switches and parallel tracks.

Table 3.1 Relationship between representative prior work and this thesis.

Work	Map information	Estimation principle	Relevance here
Quddus et al. (2007; 2003)	Road geometry and topology	Map matching taxonomy and algorithm	Establishes association cues and failure factors
Newson and Krumm (2009)	Road candidates and network transitions	Hidden Markov model	Shows benefit of sequence continuity under noise and sparsity
El Najjar and Bonnifait (2005)	Road segment geometry	Kalman-filter position observation with road-aligned covariance	Direct basis for the line-normal constraint
Fouque et al. (2008)	Road direction	Kalman-filter heading observation	Direct basis for the heading constraint and consistency testing
Tao et al. (2017)	Road-oriented information	Road-centred EKF	Supports direction-dependent road-frame uncertainty
Heirich (2016)	Track position, heading, curvature, topology	Rao-Blackwellized particle filter	Explains direction ambiguity, parallel tracks, and switches
Winter et al. (2019)	Track geometry	Map-aided train localization	Motivates explicit evaluation of cross-track accuracy
Vora et al. (2020)	Aerial imagery and LiDAR ground reflectivity	Cross-modal registration	Provides a direct basis for an orthophoto-derived planar-pose observation
Fervers et al. (2022, 2023)	Orthophotos with onboard camera and LiDAR information	Uncertainty-aware cross-view localization and tracking	Shows how pose uncertainty and registration validity can be retained
This thesis	OSM geometry and orthophoto-derived map information	ESKF pseudo-measurements	Compares output-only, coordinate, orientation, and combined-pose updates

The methods are not identical and their reported results cannot be transferred directly to the present datasets. Their common lesson is architectural: map information must be associated, assigned uncertainty, checked for consistency, and integrated in a way that matches its true information content.

3.6 Research Gap

Prior studies establish that route geometry can support localization, but three practical questions remain for the present implementation. First, it is unclear which information structure is most useful when the same OSM polyline is expressed as a heading, a scalar lateral displacement, or a stacked lateral-position and heading observation. Second, the visible benefit of snapping must be separated from the estimation benefit of changing the state and covariance. Third, performance must be interpreted together with association validity and innovation acceptance, not only with final trajectory error.

The thesis addresses these questions by holding the IMU propagation, GNSS model, initialization, sensor input, and evaluation interval constant. The primary experimental variable is the information supplied by the map: no map update, an output-only correction, an orientation constraint, a coordinate constraint, or a combined pose constraint. The comparison therefore includes GNSS/IMU fusion, conventional output-only matching, and the applicable one-step constraint formulations derived from OSM and orthophoto map sources.

4 Methodology

This chapter derives the estimation and map-constraint models used in the experiments. It begins with the complete navigation architecture and state, then describes GNSS/IMU fusion, OSM segment association, the three map observations, and the proposed one-step update. The derivation follows the implemented computation order so that every experimental mode can be traced to a stated equation.

4.1 System Architecture

The estimator operates on a chronological stream of IMU and GNSS records. IMU measurements propagate the state at high rate. When a GNSS timestamp lies between two IMU timestamps, an interpolated IMU record is inserted so that the state is first propagated exactly to the GNSS epoch. The GNSS position update is then applied if the observation is available.

For OSM-aided modes, a second correction follows the GNSS update. The current posterior position, yaw, and covariance are supplied to the segment matcher. The matcher returns a closest point, tangent, normal, and association status. One of three map-observation models converts this geometry into a residual, Jacobian, and covariance. If the observation passes its NIS test, it updates the same ESKF state and covariance. The corrected state is propagated by all subsequent IMU measurements.

The orthophoto-aided modes use exactly the same pipeline. Only the segment geometry comes from the orthophoto track digitization instead of the OSM ways (Section 5.3), so differences between the two map sources can be attributed to map geometry and not to the estimator.

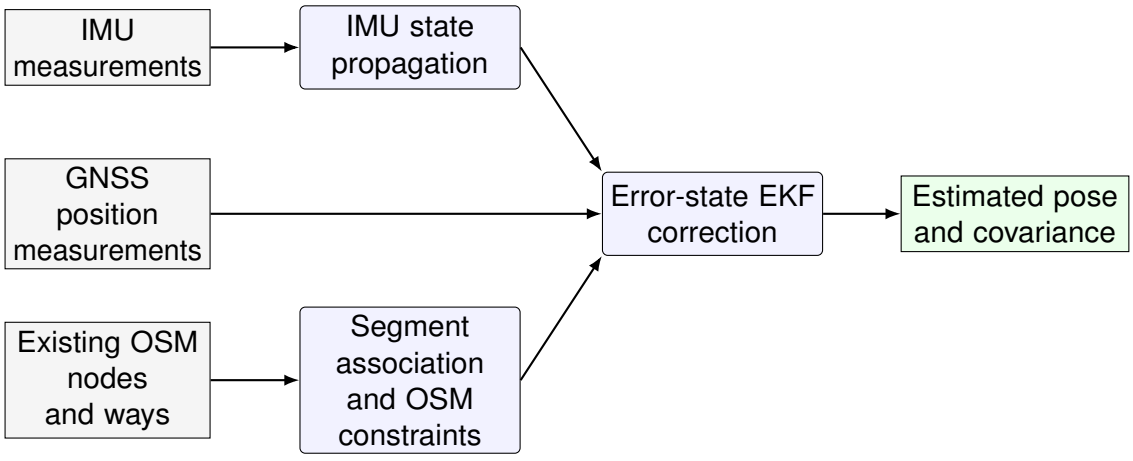


Figure 4.1 System architecture of the implemented OSM-aided estimator. The map branch uses existing OSM nodes and ways; it does not create new route geometry.

The ordering of updates is deliberate. At an epoch with valid GNSS, the map observation uses the state after GNSS correction. If no valid GNSS observation is available at an epoch, the map branch uses the IMU-propagated state. This ordering is identical for every comparison configuration.

4.2 State Space Model

Nominal and error states

The implemented estimator uses a quaternion nominal attitude and a local three-dimensional attitude error, following the error-state formulation described by Solà (2017). The nominal state is written as

$$\boldsymbol{x} = (\boldsymbol{q}_{wb}, \boldsymbol{p}_{wb}, \boldsymbol{v}_{wb}, \boldsymbol{b}_g, \boldsymbol{b}_a, \boldsymbol{g}_w)\,, \tag{4.1}$$

where $\boldsymbol{q}_{wb}$ rotates body-frame vectors into the world frame, $\boldsymbol{p}_{wb}$ and $\boldsymbol{v}_{wb}$ are world-frame position and velocity, $\boldsymbol{b}_g$ and $\boldsymbol{b}_a$ are residual IMU biases, and $\boldsymbol{g}_w$ is gravity expressed in the world frame. With four quaternion components and three components for each of the other five vectors, the nominal state has 19 values.

The world frame is an east–north–up frame realized by UTM coordinates. The GNSS receiver delivers latitude, longitude, and ellipsoidal height; these are converted to UTM zone 32N (Section 5.1), and the OSM and orthophoto maps are expressed in the same grid. Easting, northing, and height then serve as the east, north, and up axes, and $\boldsymbol{g}_w$ is initialized as $[0,\, 0,\, -9.81]^\mathsf{T}\,\mathrm{m/s^2}$. This flat-Earth approximation of a local tangent frame is adequate for the 9 km route: along the route, grid north differs from true north by the meridian convergence of 0.27 °, which is the same for GNSS, maps, and filter, and grid distances are 0.04 % shorter than true distances (scale factor 0.99961), i.e. about 0.5 m over a 60 s outage at line speed. Earth rotation is not modelled; its vertical component of about 11 °/h is constant over the recording and is absorbed by the gyroscope bias state. The body frame is the forward–left–up frame of the IMU (Section 5.1). Both frames therefore have their z-axis pointing up, and the initial attitude $\boldsymbol{q}_{wb}$ is a pure yaw rotation about this common axis.

The corresponding error state is

$$\delta\boldsymbol{x} = \begin{bmatrix}\delta\boldsymbol{\theta}^\mathsf{T} & \delta\boldsymbol{p}^\mathsf{T} & \delta\boldsymbol{v}^\mathsf{T} & \delta\boldsymbol{b}_g^\mathsf{T} & \delta\boldsymbol{b}_a^\mathsf{T} & \delta\boldsymbol{g}^\mathsf{T}\end{bmatrix}^\mathsf{T} \in \mathbb{R}^{18}. \tag{4.2}$$

The fixed ordering in Equation (4.2) is used by every Jacobian. The orientation is not updated by adding quaternion components. Instead, a small rotation vector is converted into a unit quaternion and multiplied with the nominal attitude.

Gravity is initialized as $\widehat{\boldsymbol{g}}_w = [0, 0, -9.81]^\mathsf{T}\,\mathrm{m/s^2}$. It has an initial covariance but no random-walk process noise. Its error enters the velocity-error dynamics through $\partial\dot{\delta\boldsymbol{v}}/\partial\delta\boldsymbol{g} = \boldsymbol{I}_3$, so position observations can modify gravity indirectly through cross-covariances. This is a modelling choice of the present ESKF. A deterministic local-gravity model from latitude and height would be an alternative; over the short and nearly level route considered here, the estimated gravity error remains small.

IMU measurement model

The gyroscope and accelerometer measurements are modelled as

$$\boldsymbol{\omega}_{\text{meas}} = \boldsymbol{\omega}_{\text{true}} + \boldsymbol{b}_g + \boldsymbol{n}_g, \tag{4.3}$$

$$\boldsymbol{f}_{\text{meas}} = \boldsymbol{f}_{\text{true}} + \boldsymbol{b}_a + \boldsymbol{n}_a, \tag{4.4}$$

where $\boldsymbol{n}_g$ and $\boldsymbol{n}_a$ are zero-mean measurement noise terms (Groves 2013; Titterton and Weston 2004). The static bias vector stored with the recording is removed by the data loader before filtering. Consequently, the filter states $\boldsymbol{b}_g$ and $\boldsymbol{b}_a$ represent residual bias after that preprocessing step.

Bias-corrected midpoint measurements are

$$\bar{\boldsymbol{\omega}}_k = \frac{1}{2}\left(\boldsymbol{\omega}_{\text{meas},k-1} + \boldsymbol{\omega}_{\text{meas},k}\right) - \widehat{\boldsymbol{b}}_{g,k-1}, \tag{4.5}$$

$$\bar{\boldsymbol{f}}_k = \frac{1}{2}\left(\boldsymbol{f}_{\text{meas},k-1} + \boldsymbol{f}_{\text{meas},k}\right) - \widehat{\boldsymbol{b}}_{a,k-1}. \tag{4.6}$$

With sampling interval Δt_k, the implemented nominal propagation is

$$\boldsymbol{a}_{w,k} = \boldsymbol{R}(\widehat{\boldsymbol{q}}_{wb,k-1})\bar{\boldsymbol{f}}_k + \widehat{\boldsymbol{g}}_{w,k-1}, \tag{4.7}$$

$$\widehat{\boldsymbol{p}}_k^{-} = \widehat{\boldsymbol{p}}_{k-1}^{+} + \widehat{\boldsymbol{v}}_{k-1}^{+}\Delta t_k + \frac{1}{2}\boldsymbol{a}_{w,k}\Delta t_k^2, \tag{4.8}$$

$$\widehat{\boldsymbol{v}}_k^{-} = \widehat{\boldsymbol{v}}_{k-1}^{+} + \boldsymbol{a}_{w,k}\Delta t_k, \tag{4.9}$$

$$\widehat{\boldsymbol{q}}_{wb,k}^{-} = \operatorname{norm}\!\left(\widehat{\boldsymbol{q}}_{wb,k-1}^{+} \otimes \operatorname{Exp}_q(\bar{\boldsymbol{\omega}}_k\Delta t_k)\right). \tag{4.10}$$

Here, $\otimes$ is the Hamilton quaternion product, Exp_q maps a rotation vector to a quaternion, and norm restores unit length.

Error-state dynamics and covariance propagation

The continuous error dynamics used by the implementation are

$$\delta\dot{\boldsymbol{\theta}} = -[\bar{\boldsymbol{\omega}}]_\times\delta\boldsymbol{\theta} - \delta\boldsymbol{b}_g - \boldsymbol{n}_g, \tag{4.11}$$

$$\delta\dot{\boldsymbol{p}} = \delta\boldsymbol{v}, \tag{4.12}$$

$$\delta\dot{\boldsymbol{v}} = -\boldsymbol{R}_{wb}[\bar{\boldsymbol{f}}]_\times\delta\boldsymbol{\theta} - \boldsymbol{R}_{wb}\delta\boldsymbol{b}_a + \delta\boldsymbol{g} - \boldsymbol{R}_{wb}\boldsymbol{n}_a, \tag{4.13}$$

$$\delta\dot{\boldsymbol{b}}_g = \boldsymbol{n}_{bg}, \qquad \delta\dot{\boldsymbol{b}}_a = \boldsymbol{n}_{ba}, \qquad \delta\dot{\boldsymbol{g}} = \boldsymbol{0}, \tag{4.14}$$

where $[\cdot]_\times$ denotes the skew-symmetric cross-product matrix. These equations define the continuous transition matrix $\boldsymbol{F}$ and noise mapping $\boldsymbol{G}$ in

$$\delta\dot{\boldsymbol{x}} = \boldsymbol{F}\delta\boldsymbol{x} + \boldsymbol{G}\boldsymbol{w}, \qquad \boldsymbol{w} = \begin{bmatrix}\boldsymbol{n}_g^{\mathsf{T}} & \boldsymbol{n}_a^{\mathsf{T}} & \boldsymbol{n}_{bg}^{\mathsf{T}} & \boldsymbol{n}_{ba}^{\mathsf{T}}\end{bmatrix}^{\mathsf{T}}. \tag{4.15}$$

The noise covariance is

$$\boldsymbol{Q}_c = \operatorname{diag}\!\left(\sigma_g^2\boldsymbol{I}_3, \sigma_a^2\boldsymbol{I}_3, \sigma_{bg}^2\boldsymbol{I}_3, \sigma_{ba}^2\boldsymbol{I}_3\right). \tag{4.16}$$

The implementation uses first-order discretization,

$$\boldsymbol{\Phi}_k = \boldsymbol{I}_{18} + \boldsymbol{F}_k \Delta t_k, \tag{4.17}$$

$$\boldsymbol{Q}_{d,k} = \boldsymbol{G}_k \boldsymbol{Q}_c \boldsymbol{G}_k^{\mathsf{T}} \Delta t_k, \tag{4.18}$$

$$\boldsymbol{P}_k^{-} = \boldsymbol{\Phi}_k \boldsymbol{P}_{k-1}^{+} \boldsymbol{\Phi}_k^{\mathsf{T}} + \boldsymbol{Q}_{d,k}. \tag{4.19}$$

This approximation is computationally inexpensive at the high IMU rate, but its adequacy depends on the sampling interval and the process-noise parameters.

4.3 GNSS/IMU Sensor Fusion

GNSS supplies absolute position in UTM coordinates. The recorded GNSS position is applied directly as a three-dimensional position observation; the filter state refers to the GNSS antenna. No separately derived GNSS velocity observation is used in the primary experiment. The observation model is

$$\boldsymbol{z}_{g,k} = \boldsymbol{p}_{\mathrm{GNSS},k} = \boldsymbol{p}_{wb,k} + \boldsymbol{n}_{g,k}, \quad \boldsymbol{n}_{g,k} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_{g,k}), \tag{4.20}$$

and the innovation is

$$\boldsymbol{r}_{g,k} = \boldsymbol{p}_{\mathrm{GNSS},k} - \widehat{\boldsymbol{p}}_{wb,k}^{-}. \tag{4.21}$$

The corresponding Jacobian selects only the position-error block:

$$\boldsymbol{H}_g = \begin{bmatrix} \boldsymbol{0}_3 & \boldsymbol{I}_3 & \boldsymbol{0}_3 & \boldsymbol{0}_3 & \boldsymbol{0}_3 & \boldsymbol{0}_3 \end{bmatrix}. \tag{4.22}$$

Although velocity is not observed directly, the GNSS position update can still change the estimated velocity through the position–velocity cross-covariance in $\boldsymbol{P}_k^{-}$. The implementation therefore does not construct an additional velocity measurement from consecutive GNSS positions.

The observation covariance is mode dependent. At every epoch, the receiver solution flag selects the position standard deviation for RTK fixed, RTK float, or standalone operation. This is an adaptive stochastic observation model: $\boldsymbol{R}_{g,k}$ changes when the receiver solution class changes. It is not an online-learning or innovation-adaptive model. The default three-dimensional update keeps height and vertical velocity observable. A horizontal-only configuration is reported only as a deliberately planar ablation. GNSS NIS is recorded together with its degrees of freedom and 99% chi-square threshold. Hard GNSS outlier rejection is optional and is disabled in the main baseline so that an isolated rejection cannot cause a persistent dead-reckoning lockout.

When a GNSS record is marked unavailable, the filter still produces an output at the timestamp and may process a map constraint.

Antenna height and track point

The GNSS antenna is mounted on the roof of the vehicle, whereas the maps and the evaluation reference describe the track. The filter position $\boldsymbol{p}_{wb}$ refers to the antenna. The corresponding point on the track is

$$\boldsymbol{p}_t = \boldsymbol{p}_{wb} + \boldsymbol{R}_{wb}\, \boldsymbol{t}_b, \qquad \boldsymbol{t}_b = \begin{bmatrix} 0 & 0 & -h \end{bmatrix}^{\mathsf{T}}, \tag{4.23}$$

with $\boldsymbol{t}_b$ in the forward–left–up body frame and the antenna height above the track $h = 3\,\mathrm{m}$. On canted track the vehicle rolls by up to about 5 °, and the roof antenna moves sideways relative to the track by $h \sin\phi \approx 0.26\,\mathrm{m}$. In the recording, the lateral offset between the recorded GNSS and the reference follows the estimated roll angle closely (correlation coefficient of 0.94 in magnitude), which confirms this effect.

The track point is used wherever the filter meets track geometry: the association, the closest point, and the lateral residual of the map observations are evaluated at the horizontal components of $\boldsymbol{p}_t$, and the reported position that is compared with the reference is $\boldsymbol{p}_t$. The GNSS update in Equation (4.20) still observes the antenna directly. In the map update the attitude in Equation (4.23) is treated as known, so the map Jacobians keep the form of Sections 4.5 and 4.6. A map residual therefore moves the position but does not tilt the vehicle. Modelling the offset as a lever arm inside the GNSS update instead was also tested; it reached a similar nominal accuracy, but the filter then explained map errors by tilting the vehicle, and the line-normal filter became unstable during the outages.

GNSS reacquisition after an outage

After an outage the position can be wrong by tens of metres. If the first GNSS update after the outage is applied with the propagated covariance, the large correction is distributed through the cross-covariances into attitude and biases; in the experiments this left heading errors of several tens of degrees after the end of a window. On the first GNSS epoch after an outage of duration T, with position innovation $\boldsymbol{r}_g$, the position and velocity covariance is therefore reset before the update,

$$\boldsymbol{P}_{pp} = \mathrm{diag}\big(\max(P_{pp,ii}, r_{g,i}^2)\big), \qquad \boldsymbol{P}_{vv} = \mathrm{diag}\big(\max(P_{vv,ii}, (r_{g,i}/T)^2)\big), \tag{4.24}$$

and all cross-covariances of position and velocity, with each other and with the other states, are set to zero. The first fix then corrects position and velocity only.

4.4 Map Representation

The selection of the matched segment follows the two-step map-matching literature. Candidate segments are scored by their distance to the predicted position and by the agreement of segment and vehicle direction, as in the weight-based point-to-curve algorithms reviewed by Quddus, Ochieng, and Noland (2007). The sequence of selected segments is decided with a cumulative cost over time, which is the Viterbi recursion of hidden-Markov map matching (Newson and Krumm 2009) in an online form with a limited number of hypotheses. The same selection is used by every map mode. The two-step baseline, the heading observation, the line-normal constraint, and the combined observation receive the identical segment, tangent, normal, and closest point; up to this point the information is the same, and the modes differ only in what is done with it.

OSM polyline geometry

The SQLite map already stores projected `east` and `north` values for OSM nodes. A second table stores the node order within each way. The implementation joins these tables, sorts by way identifier and node number, and connects consecutive nonidentical nodes. It therefore consumes already digitized map geometry; no route axis is traced or extracted in this work.

For segment i, let

$$\boldsymbol{s}_i = \boldsymbol{b}_i - \boldsymbol{a}_i, \qquad \ell_i^2 = \boldsymbol{s}_i^{\mathsf{T}} \boldsymbol{s}_i. \tag{4.25}$$

Segments with ℓ_i numerically equal to zero are discarded. For predicted position $\widehat{\boldsymbol{p}}_{xy}$, the finite-segment projection is

$$u_i = \mathrm{clip}\left(\frac{(\widehat{\boldsymbol{p}}_{xy} - \boldsymbol{a}_i)^{\mathsf{T}} \boldsymbol{s}_i}{\ell_i^2}, 0, 1 \right), \tag{4.26}$$

$$\boldsymbol{c}_i = \boldsymbol{a}_i + u_i \boldsymbol{s}_i, \tag{4.27}$$

$$\boldsymbol{t}_i = \frac{\boldsymbol{s}_i}{\ell_i}, \qquad \boldsymbol{n}_i = \begin{bmatrix} -t_{i,y} & t_{i,x} \end{bmatrix}^{\mathsf{T}}, \tag{4.28}$$

$$d_i = \left\| \widehat{\boldsymbol{p}}_{xy} - \boldsymbol{c}_i \right\|. \tag{4.29}$$

Bidirectional heading selection

The raw segment heading is

$$\psi_i = \mathrm{atan2}(t_{i,y}, t_{i,x}). \tag{4.30}$$

The implementation evaluates forward and reverse discrepancies,

$$\Delta\psi_{i,f} = \left| \mathrm{wrap}(\psi_i - \widehat{\psi}) \right|, \tag{4.31}$$

$$\Delta\psi_{i,r} = \left| \mathrm{wrap}(\psi_i + \pi - \widehat{\psi}) \right|, \tag{4.32}$$

$$\Delta\psi_i = \min(\Delta\psi_{i,f}, \Delta\psi_{i,r}). \tag{4.33}$$

If the reverse discrepancy is smaller, both $\boldsymbol{t}_i$ and its resulting normal are reversed. The matched heading is therefore consistent with the predicted direction of travel instead of being determined by database node order.

Uncertainty-aware association score

Distance and heading are combined in a dimensionless score. Let $\lambda_{\max}(\boldsymbol{P}_{xy})$ be the largest eigenvalue of the predicted horizontal position covariance and $P_{\psi\psi}$ the predicted yaw variance. The score is

$$\sigma_d^2 = \sigma_{d,\mathrm{map}}^2 + \lambda_{\max}(\boldsymbol{P}_{xy}), \tag{4.34}$$

$$\sigma_\psi^2 = \sigma_{\psi,\mathrm{map}}^2 + P_{\psi\psi}, \tag{4.35}$$

$$J_i = \frac{d_i^2}{\sigma_d^2} + \frac{\Delta\psi_i^2}{\sigma_\psi^2}. \tag{4.36}$$

Only candidates within a maximum Euclidean distance of 30 m are considered. The use of the largest position eigenvalue is conservative and rotationally invariant, although it does not use the full direction-dependent covariance.

Association over time

A single epoch can be misleading near switches or parallel tracks. The association therefore keeps up to five hypotheses from epoch to epoch. Each candidate segment j at epoch k receives the cumulative cost

$$C_j(k) = J_j(k) + \min_i \big(C_i(k-1) + \pi_{ij}\big), \tag{4.37}$$

where i runs over the hypotheses of the previous epoch and the transition penalty π_{ij} is zero for the same way within two segments, 0.5 per additional segment jump on the same way, and 4 for a change to another way. The candidate with the lowest cost is selected, and the costs are shifted so that the best one is zero. If a hypothesis on another way is almost as good,

$$C_{\text{alternative}} - C_{\text{best}} \leq \Delta C_{\min} = 2, \tag{4.38}$$

the association is flagged as ambiguous and no map update is applied; the alternatives are kept for the next epoch. The transition penalties favour staying on the same way, which suppresses jumps to a neighbouring track caused by a single noisy epoch.

4.5 Position Constraints

Scalar lateral-position constraint

For an accepted match, the signed cross-track displacement of the track point (Equation (4.23)) is

$$e_\perp = \boldsymbol{n}^{\mathsf{T}}(\widehat{\boldsymbol{p}}_{t,xy} - \boldsymbol{c}). \tag{4.39}$$

The map supplies the soft observation $e_\perp = 0$, giving innovation

$$r_\perp = -e_\perp. \tag{4.40}$$

The implemented Jacobian is

$$\boldsymbol{H}_\perp = \begin{bmatrix} \mathbf{0}_{1\times3} & n_x & n_y & 0 & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} \end{bmatrix}, \tag{4.41}$$

with the error-state order attitude, position, velocity, gyroscope bias, accelerometer bias, gravity, and $R_\perp = \sigma_\perp^2$. Because $\boldsymbol{H}_\perp$ has no tangent component, the update directly constrains only lateral position. Any change in along-track position, velocity, orientation, or bias arises through cross-covariance in $\boldsymbol{P}$. The constraint is the limiting case of the road-segment observation of El Najjar and Bonnifait (2005), whose covariance is elongated along the segment; for a straight segment their along-road uncertainty is infinite and only the transverse position is corrected.

4.6 Orientation Constraints

The chosen segment tangent yields the map heading

$$\psi_m = \operatorname{atan2}(t_y, t_x). \tag{4.42}$$

Yaw is extracted from the nominal rotation matrix as

$$\widehat{\psi} = \operatorname{atan2}(R_{21}, R_{11}), \tag{4.43}$$

and the residual is wrapped to $(-\pi, \pi]$:

$$r_\psi = \operatorname{wrap}(\psi_m - \widehat{\psi}). \tag{4.44}$$

Because the attitude error is applied in the body frame, $\boldsymbol{q} \leftarrow \boldsymbol{q} \otimes \delta\boldsymbol{q}(\delta\boldsymbol{\theta})$, the derivative of the world yaw with respect to $\delta\boldsymbol{\theta}$ depends on roll and pitch. It is computed by a central difference on the quaternion,

$$\frac{\partial\psi}{\partial\delta\theta_j} \approx \frac{\psi\big(\boldsymbol{q} \otimes \delta\boldsymbol{q}(\epsilon\boldsymbol{e}_j)\big) - \psi\big(\boldsymbol{q} \otimes \delta\boldsymbol{q}(-\epsilon\boldsymbol{e}_j)\big)}{2\epsilon}, \qquad \boldsymbol{H}_\psi = \begin{bmatrix} \dfrac{\partial\psi}{\partial\delta\boldsymbol{\theta}} & \boldsymbol{0}_{1\times 15} \end{bmatrix}, \tag{4.45}$$

with $\epsilon = 10^{-7}\,\mathrm{rad}$. Because the body and world z-axes both point up, it is close to $[0,\ 0,\ 1]$; over the route its components stay within $[0,\ \pm 0.09,\ 1.00]$. The second component equals the sine of the roll angle, which reaches about 5 ° on canted track in curves.

The heading observation has a constant variance $R_\psi = \sigma_\psi^2$, with the same value for both maps and at every speed. A rail vehicle is aligned with the track also when it stands still, so the map direction remains a valid observation of yaw at low speed. The value of σ_ψ is derived in Section 5.4.

For the combined constraint, the lateral-position and heading observations are stacked into one update:

$$\boldsymbol{r}_m = \begin{bmatrix} -e_\perp \\ \operatorname{wrap}(\psi_m - \widehat{\psi}) \end{bmatrix}, \qquad \boldsymbol{R}_m = \begin{bmatrix} \sigma_\perp^2 & 0 \\ 0 & \sigma_\psi^2 \end{bmatrix}. \tag{4.46}$$

The corresponding Jacobian stacks $\boldsymbol{H}_\perp$ and $\boldsymbol{H}_\psi$. Both observations originate from the same associated segment. The diagonal covariance neglects possible correlation between the map-derived lateral position and heading, so its adequacy is checked with the NIS.

4.7 Constrained Extended Kalman Filter

All GNSS and map observations use the same linearized correction routine. For innovation $\boldsymbol{r}$, Jacobian $\boldsymbol{H}$, and covariance $\boldsymbol{R}$,

$$\boldsymbol{S} = \boldsymbol{H}\boldsymbol{P}^-\boldsymbol{H}^\mathsf{T} + \boldsymbol{R}, \tag{4.47}$$

$$\mathrm{NIS} = \boldsymbol{r}^\mathsf{T}\boldsymbol{S}^{-1}\boldsymbol{r}, \tag{4.48}$$

$$\boldsymbol{K} = \boldsymbol{P}^-\boldsymbol{H}^\mathsf{T}\boldsymbol{S}^{-1}, \tag{4.49}$$

$$\delta\widehat{\boldsymbol{x}} = \boldsymbol{K}\boldsymbol{r}. \tag{4.50}$$

The predicted state covariance $\boldsymbol{P}^-$ and the observation covariance $\boldsymbol{R}$ are first combined in the innovation covariance $\boldsymbol{S}$. The NIS is then evaluated. If the observation is accepted, $\boldsymbol{S}$ determines the Kalman gain, and the gain maps the innovation into the error-state correction. Thus, $\boldsymbol{R}$ is not used to update $\boldsymbol{P}^-$ in a separate preliminary step; both covariances jointly determine the weight of the observation. The implementation solves linear systems involving $\boldsymbol{S}$ rather than forming an explicit inverse. The NIS measures how large the innovation is compared with its expected uncertainty $\boldsymbol{S}$. A map observation is rejected when its NIS exceeds the 99%

chi-square threshold for its dimension; this test is called the innovation gate. An observation that fails the gate is treated as an outlier, for example a wrong association or a map error. The one-dimensional heading and lateral observations use $\chi^2_{1,0.99} = 6.6349$; the two-dimensional combined observation uses $\chi^2_{2,0.99} = 9.2103$ (Bar-Shalom et al. 2001).

For an accepted observation, the correction is injected compactly with the error-state composition operator $\boxplus$,

$$\widehat{\boldsymbol{x}}^+ = \widehat{\boldsymbol{x}}^- \boxplus \delta\widehat{\boldsymbol{x}} = \left(\mathrm{norm}\left(\widehat{\boldsymbol{q}}_{wb}^- \otimes \mathrm{Exp}_q(\delta\widehat{\boldsymbol{\theta}})\right), \widehat{\boldsymbol{p}}^- + \delta\widehat{\boldsymbol{p}}, \widehat{\boldsymbol{v}}^- + \delta\widehat{\boldsymbol{v}}, \widehat{\boldsymbol{b}}_g^- + \delta\widehat{\boldsymbol{b}}_g, \widehat{\boldsymbol{b}}_a^- + \delta\widehat{\boldsymbol{b}}_a, \widehat{\boldsymbol{g}}^- + \delta\widehat{\boldsymbol{g}}\right). \tag{4.51}$$

The quaternion component is composed multiplicatively, whereas the remaining Euclidean components are corrected additively (Solà 2017).

The covariance uses the Joseph form,

$$\boldsymbol{P}' = (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\boldsymbol{P}^-(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})^\mathsf{T} + \boldsymbol{K}\boldsymbol{R}\boldsymbol{K}^\mathsf{T}. \tag{4.52}$$

After attitude-error injection, the local error coordinates are reset using

$$\boldsymbol{J}_{r,\theta} \simeq \boldsymbol{I}_3 - \frac{1}{2}[\delta\widehat{\boldsymbol{\theta}}]_\times, \qquad \boldsymbol{P}^+ = \boldsymbol{J}_r\boldsymbol{P}'\boldsymbol{J}_r^\mathsf{T}. \tag{4.53}$$

The resulting covariance is symmetrized numerically. These steps preserve the local interpretation of the attitude error and reduce avoidable numerical loss of symmetry.

4.8 Proposed One-Step Map Integration

The proposed processing sequence at output epoch k is:

1. propagate the nominal state and covariance through every IMU sample up to timestamp t_k;
2. if GNSS is available, form the position residual and apply the GNSS correction, after the covariance reset of Equation (4.24) on the first epoch after an outage;
3. compute the track point $\boldsymbol{p}_t$ (Equation (4.23)) and query every map segment using its horizontal position, the yaw, the position covariance, and the yaw variance;
4. reject the association if the best segment is farther than the maximum distance or if another way falls inside the ambiguity margin;
5. construct exactly one selected map observation: heading, lateral position, or combined lateral position and heading;
6. reject the map observation if its NIS exceeds the configured chi-square threshold; and
7. otherwise inject the map correction and propagate its posterior state and covariance into the next epoch.

This sequence is one-step because the map update changes the recursive estimator. In compact form,

$$(\widehat{\boldsymbol{x}}_k^+, \boldsymbol{P}_k^+) = \mathcal{U}_m\big(\mathcal{U}_g(\widehat{\boldsymbol{x}}_k^-, \boldsymbol{P}_k^-), \mathcal{M}\big), \tag{4.54}$$

where $\mathcal{U}_g$ and $\mathcal{U}_m$ denote the GNSS and map updates.

The two-step baseline uses the same segment association but replaces only the reported horizontal output:

$$\widehat{\boldsymbol{p}}_{\mathrm{out},xy} = \boldsymbol{c}, \qquad (\widehat{\boldsymbol{x}}_k^+, \boldsymbol{P}_k^+) = (\widehat{\boldsymbol{x}}_{g,k}^+, \boldsymbol{P}_{g,k}^+). \tag{4.55}$$

At the next epoch, two-step propagation therefore begins from the GNSS/IMU state, not from the displayed closest point. Comparing Equations (4.54) and (4.55) isolates the value of filter feedback from the cosmetic effect of forcing an output onto the map.

5 Implementation

This chapter maps the methodology of Chapter 4 to the implemented software. It documents the actual data fields, database operations, parameter values, state initialization, update ordering, and exported diagnostics. This separation is important: the Methodology chapter defines the general models, whereas this chapter states which choices are present in the current code.

5.1 Dataset Description

Track-constrained recording

The primary experiment uses recording date 30 January 2025, set 13. The data consist of one IMU file and one GNSS file in MATLAB format. The IMU file contains 234,159 accelerometer records and the same number of gyroscope records at a declared sampling rate of 300 Hz. Their timestamps cover approximately 780.45 s. The GNSS file contains 780 position records at 1 Hz. Each position record contains timestamp, latitude, longitude, and altitude. The recorded trajectory is predominantly track constrained and contains sufficiently fast motion for map heading to be informative over much of the route.

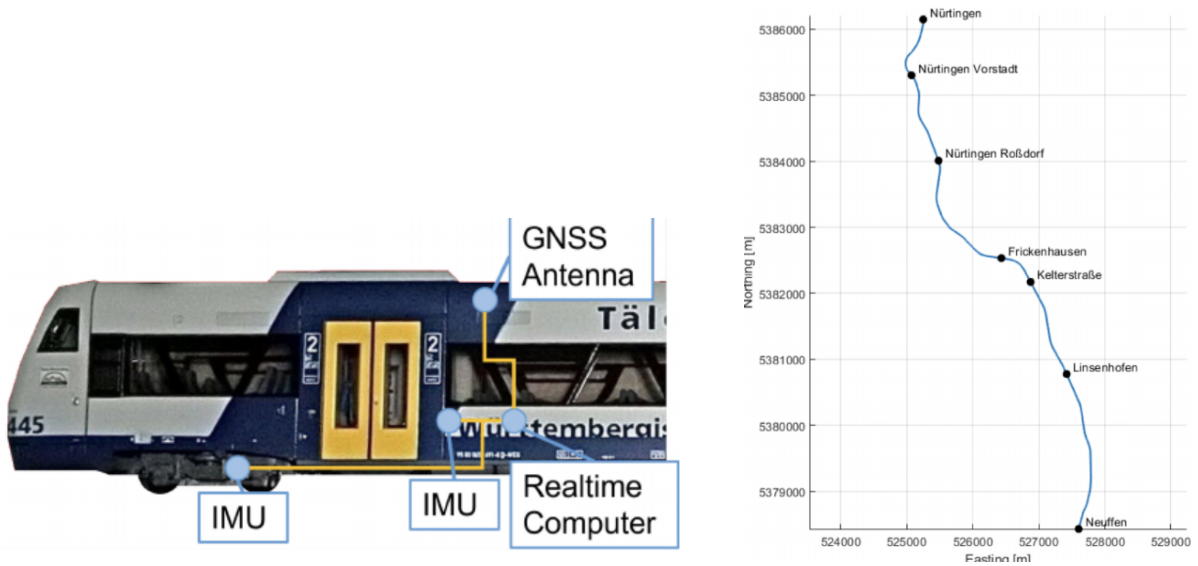
The recording comes from the measurement campaign described by Frolow et al. (2026). The measurement vehicle carried one IMU on the car body and a second IMU on the bogie; the recording used here contains one IMU stream from the car body. Consequently, this thesis uses and reports one 300 Hz IMU input and does not claim fusion of two IMUs. The sensor arrangement and the seven-station test route are shown in Figure 5.1.

The accelerometer and gyroscope arrays have the form

$$\boldsymbol{D}_{\mathrm{IMU}} = \begin{bmatrix} t_1 & m_{x,1} & m_{y,1} & m_{z,1} \\ \vdots & \vdots & \vdots & \vdots \\ t_N & m_{x,N} & m_{y,N} & m_{z,N} \end{bmatrix}. \tag{5.1}$$

Before filtering, the loader verifies that accelerometer and gyroscope shapes and timestamps agree and that every timestamp sequence is strictly increasing. Gyroscope values are converted from degrees per second to radians per second.

The IMU metadata provide a six-component bias vector and six axis-orientation signs, ordered as accelerometer x, y, z followed by gyroscope x, y, z. The bias vector stored with the recording is subtracted and the six axis-orientation signs are applied, which gives a forward–right–down frame. A rotation of 180° about the forward axis, i.e. a sign change of the y and z components, then gives the forward–left–up body frame used by the filter, whose z-axis points up like that of the east–north–up world frame. The frame was checked against the data: at standstill the accelerometer measures $+9.80\,\mathrm{m/s^2}$ along z; the z-gyroscope follows the heading rate of the



(a) Sensor locations in the measurement vehicle. The positioning filter uses the car-body IMU; the bogie IMU is not part of the positioning input used in this thesis.

(b) Approximately 9 km test route between Nürtingen and Neuffen with seven stations.

Figure 5.1 Measurement vehicle and test route (Frolow et al. 2026, figures licensed under CC BY 4.0).

GNSS track (correlation 0.995), and the x-accelerometer follows the change of GNSS speed (correlation 0.98).

GNSS latitude and longitude are converted from WGS84 to a northern UTM zone selected from the median longitude. The implemented Transverse Mercator conversion uses the WGS84 semi-major axis $a = 6378137\,\mathrm{m}$, flattening $f = 1/298.257223563$, scale factor $k_0 = 0.9996$, and 500 000 m false easting; for the recording it agrees with the PROJ library (EPSG:32632) to better than 1 mm. The result is

$$\boldsymbol{z}_{g,k} = \begin{bmatrix} E_k & N_k & h_k \end{bmatrix}^{\mathsf{T}}. \tag{5.2}$$

When no per-epoch GNSS covariance is available, the loader assigns the horizontal standard deviation according to the recorded solution mode,

$$\sigma_{p,k} = \begin{cases} 0.10\,\mathrm{m}, & \text{RTK fixed}, \\ 0.50\,\mathrm{m}, & \text{RTK float}, \\ 5.00\,\mathrm{m}, & \text{standalone or fallback}. \end{cases} \tag{5.3}$$

The vertical standard deviation is the larger of twice the horizontal value and 3 m. The covariance therefore changes when the receiver mode changes. These values are the adaptive stochastic model determined for this measurement campaign by Frolow et al. (2026): mode 4 is RTK fixed (0.10 m), mode 5 RTK float (0.50 m). The 5 m value is a conservative fallback for an unknown or standalone mode and is not used by set 13. An RTK-fixed solution means that the receiver has resolved the integer ambiguities of the carrier-phase observations. It normally indicates a more precise solution than RTK float, in which the ambiguities remain real-valued estimates, but the flag is not itself an accuracy guarantee. Under this interpretation, set 13 contains 779 RTK-fixed records and one RTK-float record. Consequently, changing from a constant 0.10 m horizontal covariance to the mode-dependent covariance affects essentially one update in this recording. Retaining the mode-dependent logic is nevertheless preferable to discarding an available quality flag. The default experiment uses all three GNSS position components but no

separately derived GNSS velocity observation. The horizontal-only setting is kept solely for an explicit planar ablation.

The IMU and GNSS streams are merged chronologically. If GNSS time t_g lies between IMU times $t_i < t_g < t_{i+1}$, the loader constructs an interpolated IMU sample

$$\boldsymbol{m}(t_g) = (1-\alpha)\boldsymbol{m}(t_i) + \alpha\boldsymbol{m}(t_{i+1}), \qquad \alpha = \frac{t_g - t_i}{t_{i+1} - t_i}. \tag{5.4}$$

This sample is emitted immediately before the GNSS record. The filter therefore never applies a position measurement to a state whose timestamp differs from the observation timestamp. The interpolated value is not an additional IMU measurement and does not increase the amount of sensor information. It is a numerical estimate of the IMU input over the fractional interval between the last original sample and the GNSS epoch. Every original IMU sample remains in the chronological stream.

The evaluation reference is a polyline of the track centre line with 8150 points at about 1 m spacing, digitized from official true orthophotos with a ground sampling distance of $0.20\,\mathrm{m} \times 0.20\,\mathrm{m}$ per pixel. It is loaded through a separate evaluation component and is not passed to the ESKF, the OSM matcher, or any constraint generator. This prevents the reference from becoming an unreported filter measurement. Epochs that extrapolate before its first node or after its last node are excluded because proximity to a polyline endpoint does not establish reference coverage; the reference ends about 0.9 km (103 s) before the end of the recording.

Table 5.1 Data sources and their roles in the current implementation.

Source	Information used	Role
Track IMU file	Time, acceleration, angular rate, metadata	High-rate state and covariance propagation
Track GNSS file	Time, WGS84 position, fix mode	Absolute position observations and initialization
`track.db`	Projected OSM nodes and ordered ways	Association and map-constraint generation
Evaluation reference	Projected reference polyline	Evaluation only; never supplied to the filter
Orthophoto track digitization	Track polyline digitized from true orthophotos (about 5 m spacing)	Orthophoto map source for the constraints; never used as evaluation truth
Withheld GNSS	Recorded RTK positions inside outage windows	Evaluation truth in the outage experiment only

5.2 OpenStreetMap Processing

The OSM track map is stored in the SQLite database `track.db`. It holds 439 nodes on 16 OSM ways along the route, with coordinates already projected to UTM zone 32N; the newest node edit dates from 13 July 2026. The map data are © OpenStreetMap contributors and available under the Open Database License.

Database schema and map loading

The implementation reads the SQLite map database in read-only mode. The two tables used by the matcher are:

- `map(node_id, east, north, osm_timestamp)`, containing the projected position of every stored OSM node; and
- `node_order(node_id, way_id, node_number)`, containing the ordered membership of nodes in each way.

The database contains 439 map nodes, 454 ordered node references, and 16 ways. Joining both tables and connecting consecutive nodes produces 438 nondegenerate segments. The database already contains easting and northing; no geographic coordinate conversion and no manual digitization occur in the map loader.

The join is ordered by `way_id` and `node_number`. When the way identifier changes, a new polyline begins. For two consecutive points $\boldsymbol{a}_i$ and $\boldsymbol{b}_i$, the implementation stores

$$(\boldsymbol{a}_i, \boldsymbol{b}_i, \boldsymbol{s}_i, \ell_i^2, \boldsymbol{t}_i, \mathrm{way}_i, \mathrm{index}_i). \tag{5.5}$$

Adjacent points separated by no more than $10^{-6}\,\mathrm{m}$ are treated as duplicates and do not produce a segment. Endpoint arrays, direction arrays, and identifiers are stored as NumPy arrays so that projection onto all segments can be evaluated without a Python loop.

Vectorized association

For a query position, the matcher evaluates the projection fraction, closest point, squared distance, raw tangent heading, forward heading error, and reverse heading error for every segment. It then computes the score in Equation (4.36) and sorts candidate indices by this score.

The nominal association parameters are

$$d_{\max} = 30\,\mathrm{m}, \qquad \sigma_{d,\mathrm{map}} = 2\,\mathrm{m}, \qquad \sigma_{\psi,\mathrm{map}} = 10^\circ, \qquad \Delta J_{\min} = 2. \tag{5.6}$$

The ten best candidates are passed to the association over time (Equation (4.37)), which keeps up to five of them as hypotheses. Segments on the same way are ignored in the ambiguity comparison because neighbouring segments of one polyline naturally have similar scores near their common node. If a hypothesis on another way lies within the cost margin, the association is marked ambiguous.

These four quantities serve different purposes and are not all measurement uncertainties. The distance limit $d_{\max}$ is a permissive search guard that prevents a distant segment from being accepted when the predicted position is far outside the mapped route. The scales $\sigma_{d,\mathrm{map}}$ and $\sigma_{\psi,\mathrm{map}}$ only normalize distance and heading within the association score; they do not enter the ESKF observation covariance. Their values balance the two score terms while retaining candidate segments under the expected GNSS prediction error. Finally, $\Delta J_{\min}$ requires the best candidate to be separated from a competing way by at least two score units. These are frozen engineering parameters of the association stage, not values fitted to minimize the reported trajectory RMSE. Their effect is reported through association availability and ambiguity counts.

The match object contains way identifier, segment index, closest point, tangent, normal, selected heading, Euclidean distance, signed lateral error, score, and ambiguity flag. These values are

shared by every constraint model, ensuring that differences among heading, lateral, and pose experiments are not caused by different matchers.

5.3 Orthophoto Processing

The orthophoto map is a track polyline digitized manually from official true orthophotos with a ground sampling distance of 0.20 m, i.e. one pixel covers 0.20 m$\times$0.20 m on the ground (1730 nodes, median spacing 4.7 m) (Frolow et al. 2026; Landesamt fuer Geoinformation und Landentwicklung Baden-Wuerttemberg 2026). It is used only as map input to the filter, and only in the outage experiment, where it is evaluated against withheld GNSS (Section 6.2). It is therefore never compared with the digitized evaluation reference, which was digitized from the same orthophotos.

Three processing steps turn the digitization into a map that the existing constraint models can use unchanged.

Lateral offset. The map observations refer to the track point on the track centre line (Equation (4.23)). The orthophoto digitization follows one rail, at a nearly constant lateral distance of about 0.7 m from the centre line, which corresponds to half the standard gauge of 1.435 m. Used unchanged, the map would pull every lateral update 0.7 m to the side. Every vertex is therefore shifted by 0.7175 m towards the centre line, along the mean normal of its adjacent segments. The offset follows from the track geometry and is not estimated from the data.

Gaps. 48 consecutive nodes are more than 10 m apart (up to 155 m), for example where forest shadows hide the track in the orthophotos. Such chords are not track geometry on curves: at the tightest radius of the route (188 m), a 10 m chord deviates by less than 7 cm from the arc, a 20 m chord already by 27 cm. Chords longer than 10 m are therefore removed, and the polyline is split into separate pieces at each gap. In total, about 1.06 km of the 8.96 km digitization is gap.

Association near gaps. Without a further test, a position inside a gap would be associated with the end vertex of the neighbouring piece. For the orthophoto map, an association is therefore accepted only if the residual is perpendicular to the segment: the along-segment component of the residual must be below 1 m. Positions in gaps then receive no map update. With this rule, the map is available at 692 of the 780 GNSS epochs, and the distance of the recorded GNSS to the associated segment has a median of 0.13 m and an RMSE of 0.26 m. The corresponding OSM values are 0.35 m and 0.54 m.

The lateral standard deviation of the orthophoto map is set to one pixel, $\sigma_{\mathrm{lat}} = 0.20\,\mathrm{m}$, before any filter run. The OSM heading, lateral, and combined constraint models, gates, and association scoring are reused without change; only the geometry source differs. A pose observation from on-board imagery registered to the orthophoto is not possible with the recording, because it contains no camera or LiDAR stream.

5.4 Constraint Generation

Common constraint interface

Every map model returns the same data structure:

$$\mathcal{C} = (\text{name}, \boldsymbol{r}, \boldsymbol{H}, \boldsymbol{R}, \gamma), \tag{5.7}$$

where γ is the NIS threshold. All residuals are formed at the track point of Equation (4.23), 3 m below the antenna. This design keeps geometry association separate from filter correction. The constraint generator has no direct access to the covariance-update code, while the ESKF does not need to know whether a residual came from heading, lateral displacement, or their combined observation.

Heading mode

The heading model returns no constraint if the match is ambiguous. Otherwise it uses the wrapped heading residual from Equation (4.44) and the numerical yaw Jacobian of Equation (4.45), with

$$\sigma_\psi = 1^\circ, \qquad \gamma = 6.6349. \tag{5.8}$$

The value is chosen in the same way as $\sigma_\perp$ below. At the epochs where the train moves faster than 2 m/s, the direction of the matched segment differs from the direction of the smoothed GNSS track by 0.87° RMS for OSM and 0.66° RMS for the orthophoto map. The RMS is used rather than the median (0.36° and 0.39°) because in curves the direction of a straight chord differs from the track tangent by up to about 2°; a value at the median would make the gate reject these observations. Both RMS values are rounded up to 1°. A heading standard deviation that grows at low speed, as often used for road vehicles, was also tested. It weakened the heading in the nominal experiment (0.64° instead of 0.49° RMSE) and let the combined filter lose its heading after the outages, when the first GNSS update after an outage corrects a large along-track error. A rail vehicle stays aligned with the track at any speed, so the constant value is used.

Lateral mode

The lateral model uses the negative signed displacement as a scalar innovation and places the segment normal in the horizontal position columns of the Jacobian. Its configured standard deviation and gate are

$$\sigma_\perp = 0.35\,\text{m}, \qquad R_\perp = \sigma_\perp^2 = 0.1225\,\text{m}^2, \qquad \gamma = 6.6349. \tag{5.9}$$

The standard deviation tells the filter how far the OSM line may lie from the real track. OSM ways are drawn by volunteers and deviate from the track by some decimetres. The recorded RTK GNSS positions show where the train actually was; their median distance to the associated OSM segment is 0.35 m. A value of 0.35 m therefore matches the typical OSM deviation in this area. A smaller value would make the filter trust the map too much, a larger one would let it ignore the map. The value was fixed before the filter modes were compared. It is suppressed when the association is ambiguous.

Combined lateral-position and heading mode

The combined mode stacks the scalar lateral-position residual and the scalar heading residual. It uses the same $\sigma_\perp$ and σ_ψ as the individual modes. The two-dimensional observation is rejected for an ambiguous association or when its NIS exceeds 9.2103, the 99% chi-square threshold for two degrees of freedom.

Table 5.2 Implemented OSM constraint configurations.

Constraint	Dimension	Directly observed components	Principal safeguards
Heading	1	Yaw error	Bidirectional tangent, ambiguity flag, 1D NIS
Lateral	1	Position along segment normal	Distance limit, ambiguity flag, 1D NIS
Combined	2	Lateral position and yaw	Distance limit, ambiguity flag, 2D NIS

5.5 ESKF Implementation

Pre-filter position and motion initialization

The IMU supplies the high-rate increments used after initialization, but it cannot determine an absolute UTM position from its own measurements. In the implemented warm start, the time origin is taken from the first IMU epoch, whereas the first GNSS position initializes the nominal global position, $\widehat{\boldsymbol{p}}_0 = \boldsymbol{z}^p_{g,0}$. Before the recursive propagation and correction loop starts, up to 12 GNSS *position* fixes are fitted with a constant-velocity line solely to initialize velocity and horizontal direction. For relative times $\tau_i = t_i - t_0$, the implemented initialization is

$$(\widehat{\boldsymbol{v}}_0, \widehat{\boldsymbol{c}}) = \arg\min_{\boldsymbol{v},\boldsymbol{c}} \sum_{i=0}^{n-1} \left\| \boldsymbol{z}^p_{g,i} - (\boldsymbol{c} + \boldsymbol{v}\tau_i) \right\|^2 . \tag{5.10}$$

The fitted intercept $\widehat{\boldsymbol{c}}$ is not used as the initial position; the first recorded GNSS position remains $\widehat{\boldsymbol{p}}_0$. If the fitted horizontal speed is at least 0.2 m/s, the initial yaw is

$$\widehat{\psi}_0 = \operatorname{atan2}(\widehat{v}_{N,0}, \widehat{v}_{E,0}); \tag{5.11}$$

otherwise yaw and velocity are initialized to zero. This approach is less sensitive to 1 Hz GNSS jitter than differencing only the first two positions, but it assumes that the initial interval is sufficiently straight. This calculation is an offline warm start and is not a GNSS velocity observation in the recursive filter. During the filter run, GNSS supplies position only. Velocity then changes through IMU propagation and through the position–velocity cross-covariance of a position correction. A strictly causal implementation would require a preceding initialization interval or would initialize velocity with larger uncertainty and exclude the convergence interval from evaluation.

The matrix $\boldsymbol{P}_0$ is the initial error covariance, not an additional state or measurement. In the state order of Equation (4.2), its block-diagonal initialization is

$$\begin{aligned} \boldsymbol{P}_0 = \operatorname{blkdiag}\big(&(10^\circ)^2 \boldsymbol{I}_3,\ \boldsymbol{R}_{g,0},\ (5\,\mathrm{m/s})^2 \boldsymbol{I}_3, \\ &(2\,^\circ/\mathrm{s})^2 \boldsymbol{I}_3,\ (0.5\,\mathrm{m/s^2})^2 \boldsymbol{I}_3,\ (0.1\,\mathrm{m/s^2})^2 \boldsymbol{I}_3 \big). \end{aligned} \tag{5.12}$$

Propagation and process noise

The nominal state uses the midpoint propagation given in Equation (4.10). An interval must be positive and no larger than 0.25 s; a larger gap is treated as a data error. The default continuous-time noise parameters are

$$\sigma_g = 0.01, \quad \sigma_a = 0.10, \quad \sigma_{bg} = 10^{-4}, \quad \sigma_{ba} = 10^{-3}, \tag{5.13}$$

in rad/s, m/s^2, rad/s^2, and m/s^3, each per $\sqrt{\mathrm{Hz}}$. They are engineering values that were fixed before the experiments and kept identical for every run; they were not calibrated from a sensor specification.

Correction, injection, and numerical safeguards

GNSS and OSM constraints call one linear update function. The function computes $\boldsymbol{S}$, solves $\boldsymbol{S}\boldsymbol{y} = \boldsymbol{r}$ for NIS, and solves a second system for the Kalman gain. It never inverts the full 18-dimensional covariance. If the NIS is finite and below the threshold, the error state is injected and the Joseph covariance update and attitude-reset Jacobian are applied.

The quaternion is normalized after both propagation and correction. Every propagated and corrected covariance is checked for finite values and then symmetrized:

$$\boldsymbol{P} \leftarrow \frac{1}{2}(\boldsymbol{P} + \boldsymbol{P}^{\mathsf{T}}). \tag{5.14}$$

These operations do not guarantee statistical consistency, but they prevent avoidable numerical asymmetry and invalid quaternion norm.

Table 5.3 Principal estimator implementation choices.

Component	Implemented choice	Purpose or consequence
Attitude	Scalar-first nominal quaternion; three-component error	Unit-norm attitude with a minimal local correction (Solà 2017)
IMU integration	Midpoint measurement	Uses consecutive samples in the nominal propagation
Covariance propagation	First-order transition	Efficient at high rate; approximation depends on Δt
Covariance correction	Joseph form and reset Jacobian	Numerical stability and correct error-coordinate reset
GNSS update	Three-dimensional position by default; optional planar ablation	Stabilizes height and vertical velocity while retaining a controlled horizontal-only comparison
Map gate	99% chi-square NIS threshold	Rejects innovations inconsistent with modelled uncertainty
Association	Distance, bidirectional heading, covariance, ambiguity margin	Avoids unconditional nearest-segment updates

5.6 Software Framework

Module organization

The implementation separates data access, geometry, estimation, experiment control, and evaluation. The principal responsibilities are listed in Table 5.4.

Table 5.4 Principal software components.

Component	Responsibility
Data loader	Validation, unit conversion, WGS84-to-UTM conversion, bias preprocessing, and chronological sensor iteration
ESKF state and quaternion utilities	Nominal-state storage, quaternion products, rotation-vector conversion, and rotation matrices
GNSS/IMU ESKF	Initialization, propagation, GNSS correction, generic map correction, error injection, and covariance handling
Track database matcher	Read-only SQLite loading, segment construction, vectorized projection, scoring, direction selection, and ambiguity test
Map constraint models	Heading, lateral-position, and combined residuals, Jacobians, covariances, and NIS thresholds
Orthophoto polyline loader	Reading the orthophoto digitization, lateral offset, gap splitting, and perpendicular-residual association test
Trajectory metrics	Heading error against the reference tangent, outage windows, chainage, along-track and cross-track errors
Experiment runner	Mode selection, update ordering, and per-epoch diagnostics
Reference evaluation	Point-to-polyline distance, common coverage mask, aggregate statistics, plots, and CSV output

Experiment modes

The comparison interface distinguishes the following configurations:

Table 5.5 Software modes relevant to the map-constraint comparison.

ID	Software mode	Behaviour
B0	`gnss_imu`	Applies GNSS position corrections and no map update
B1	`two_step`	Applies GNSS internally and replaces only the reported horizontal position by the matched OSM point
C1	`one_step_map_heading`	Applies GNSS and a scalar map yaw correction
C2	`one_step_cross_track`	Applies GNSS and a scalar map cross-track correction
C3	`one_step_combined`	Applies GNSS and a stacked map lateral-position and yaw correction

The map source is selected with `-map-source osm` or `-map-source orthophoto`; GNSS outages are declared with repeated `-gnss-outage START:END` arguments. Every evaluated mode creates a fresh filter instance and receives identical GNSS/IMU input. At each GNSS output epoch, the runner records predicted position, reported position, GNSS NIS and acceptance, map NIS and acceptance, association validity, and estimated heading. Output-only mode B1 records a map acceptance when a valid association supplies the displayed closest point, but no map NIS is defined because no Kalman update occurs.

Evaluation output and reproducibility

The evaluation component projects every estimated horizontal position onto a separate reference polyline. A common coverage mask is computed once from raw GNSS positions and then reused for all modes. If d_{ref} is the distance to the reference, the mask is

$$\mathcal{I} = \{k \mid d_{\mathrm{ref}}(\boldsymbol{z}_{g,k}) \leq 20\,\mathrm{m}\}\,. \tag{5.15}$$

Computing a separate mask from every estimate would unfairly remove the largest errors of a poor method. The exported summaries contain sample count, mean, median, RMSE, 95th percentile, and maximum reference distance. A second CSV stores per-epoch positions and diagnostic flags for inspection.

The matcher and constraint tests use a synthetic straight-track database to verify projection, Jacobian dimensions and values, lateral state correction, finite covariance, and reference-polyline distance. These tests confirm basic implementation behaviour; they do not establish real-world accuracy. Final reproducibility additionally requires archiving the selected dataset interval, software revision, process noise, initial covariance, map parameters, fixed GNSS input selection, and evaluation mask.

6 Experimental Setup

6.1 Test Environment

All experiments use the recording of 30 January 2025 (set 13) from the ConMoRAIL campaign: 780 GNSS epochs at 1 Hz, 779 of them RTK fixed and one RTK float, and 300 Hz IMU data over about 13 min and 8.9 km. All configurations use the same data interval, coordinate frame, initialization, IMU propagation, GNSS measurements, and evaluation mask. The experimental variable is the information supplied by the map. Two experiments are carried out:

Nominal. All GNSS observations are used, and the OSM constraints are evaluated against the digitized evaluation reference. This shows the effect of the map when the GNSS/IMU solution is already accurate to about 0.1 m.

Outage. GNSS is withheld in five predefined 60 s windows. The filter then relies on the IMU and the map, which isolates the map's role as a sensor. The OSM and the orthophoto constraints are compared, and all errors are measured against the withheld GNSS.

The orthophoto map is used only in the outage experiment. In the nominal experiment it would be evaluated against a reference digitized from the same orthophotos, which would not be an independent test.

6.2 Reference Data

The nominal experiment uses the evaluation reference: a polyline of the track centre line, manually digitized from the 0.20 m true orthophotos at about 1 m spacing. It is compared with the track point of the filter (Equation (4.23)). It is never passed to the filter, the matcher, or any constraint generator. Cross-track and heading errors of the nominal experiment are measured against it.

The outage experiment uses the recorded RTK GNSS positions that were withheld from the filter. They are independent of both map sources and, being RTK fixed, accurate to a few centimetres, which is far below the drift being measured. Because the GNSS positions are 1 s apart (about 20 m at full speed), a smooth path is fitted through them with a cubic spline over arc length; standstill positions closer than 0.3 m to the previous one are removed first. This GNSS path agrees with the digitized evaluation reference to a median of 0.04 m, the same as the raw GNSS.

6.3 Evaluation Metrics

The results use a few terms that are defined here. An *epoch* is one GNSS measurement time. The recording has one epoch per second, 780 in total, and every method is evaluated at the

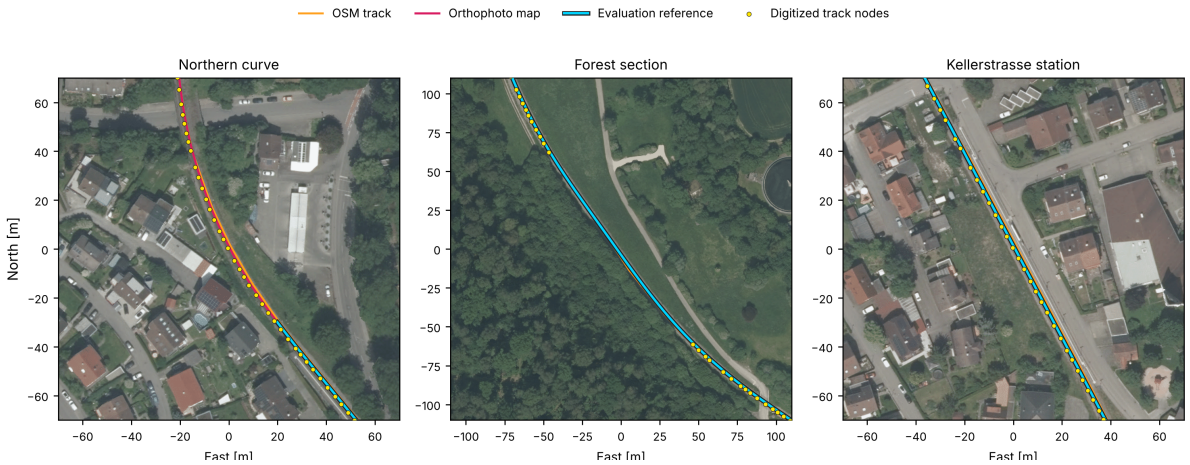


Figure 6.1 Examples from the official DOP20 true orthophoto at 0.20 m ground sampling distance. The evaluation reference and the orthophoto-derived track nodes are overlaid in their common projected coordinate frame. The background imagery documents the dataset and the manual digitization; it is not supplied directly to the ESKF. The evaluation reference ends in the northern curve, about 100 s before the end of the recording; the forest section shows a gap of the orthophoto map where trees cover the track. Orthophoto: LGL Baden-Württemberg, Datenlizenz Deutschland – Namensnennung – Version 2.0.

same epochs. The *error* of an epoch is the distance between the estimated position and the reference at that time. Its *cross-track* part is the sideways distance from the track, its *along-track* part the distance ahead of or behind the true position along the track. The errors of many epochs are summarized by the following statistics. The *RMSE* is the square root of the mean squared error and weights large errors more than the mean does. The *median* is the error that half of the epochs stay below, and the *95th percentile* the error that 95% of the epochs stay below. The *share of time within* a distance, for example within 5 cm, is the percentage of epochs whose error is smaller than that distance. A *95% bootstrap interval* gives the range of RMSE values that similar data would plausibly produce; it is obtained by recomputing the RMSE many times from randomly resampled blocks of consecutive epochs.

For a time-synchronized planar reference $\boldsymbol{p}_{\mathrm{ref},k}$, the horizontal position RMSE is

$$\mathrm{RMSE}_{xy} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} \left\| \widehat{\boldsymbol{p}}_{xy,k} - \boldsymbol{p}_{\mathrm{ref},xy,k} \right\|^2}. \tag{6.1}$$

Mean, median, 95th-percentile, and maximum horizontal errors are also reported because isolated association failures can dominate RMSE. Where the reference tangent is available, position error is decomposed into along-track and cross-track components. The signed perpendicular bias is reported using the stored direction of the reference polyline; positive values lie to its left. Because consecutive one-second epochs are correlated, RMSE uncertainty is estimated with a moving-block bootstrap rather than an independent-sample formula. Evaluation coverage is fixed from the recorded GNSS trajectory and then reused for every method. Positions beyond either global endpoint of the digitized reference are excluded; otherwise a short along-track extrapolation could be misreported as a small perpendicular error.

Heading error

Orientation accuracy is evaluated as the wrapped difference between the estimated yaw $\widehat{\psi}_k$ and the direction of the evaluation reference at the closest reference segment,

$$e_{\psi,k} = \mathrm{wrap}\Big(\widehat{\psi}_k - \mathrm{atan2}(t_{N,k}, t_{E,k})\Big), \tag{6.2}$$

where $\boldsymbol{t}_k$ is the unit tangent of that segment, flipped to the recorded direction of travel. Only the sign is taken from GNSS; the direction itself is pure reference geometry. For a rail vehicle the body yaw follows the track tangent, apart from small effects of bogie geometry. Epochs with a recorded speed below 2 m/s are excluded because yaw is weakly observable at standstill. The metric is used for evaluation only.

Outage metrics

During an outage, the error of the estimate is measured against the GNSS path and the withheld GNSS position $\boldsymbol{z}_{g,k}$ of the same epoch. Since GNSS measures the antenna, $\widehat{\boldsymbol{p}}_{xy,k}$ here is the estimated antenna position $\widehat{\boldsymbol{p}}_{wb}$, not the track point:

$$e_{\mathrm{hor},k} = \|\widehat{\boldsymbol{p}}_{xy,k} - \boldsymbol{z}_{g,k}\|, \tag{6.3}$$

$$e_{\mathrm{along},k} = s(\widehat{\boldsymbol{p}}_{xy,k}) - s(\boldsymbol{z}_{g,k}), \tag{6.4}$$

$$e_{\mathrm{cross},k} = d_{\mathrm{path}}(\widehat{\boldsymbol{p}}_{xy,k}), \tag{6.5}$$

where $s(\cdot)$ is the chainage (arc length) of the closest point on the GNSS path and d_{path} the distance to it. The cross-track error tells how far the estimate is beside the track, the along-track error how far it is ahead of or behind the vehicle. Projecting the error onto the local normal is not used: after a long along-track drift on a curve it would report along-track error as cross-track error. The heading error during outages is computed as in Equation (6.2), with the direction of the GNSS path instead of the reference. RMSE values are reported per window and over all outage epochs.

Estimator diagnostics

Estimator behaviour is described by valid-association rate, NIS distribution, rejection rate, accepted-update count, and orthophoto-registration validity where applicable. These diagnostics distinguish a consistently informative method from one that appears accurate only because most map observations were rejected.

For every complete run, the output also records height, vertical velocity, horizontal speed, the minimum covariance eigenvalue, and the maximum covariance symmetry residual at each evaluation epoch. These diagnostics expose an unobserved vertical state and verify that numerical covariance handling remains finite, symmetric, and positive semidefinite.

The propagation, GNSS, heading, lateral-position, and combined-observation Jacobians are checked against finite differences in the unit tests. The IMU process-noise values are engineering values and are kept identical in all runs. The median GNSS NIS and the proportion above the corresponding 99% chi-square threshold are reported. A filter with grossly inconsistent NIS is not used for substantive comparison even if one trajectory RMSE happens to be small.

6.4 Baseline Approaches

Two baselines are used. The *GNSS/IMU ESKF* (B0) applies no map information and shows what the sensors alone achieve. Inside the outage windows it receives no GNSS and runs on the IMU alone, so it is labelled *IMU only (GNSS withheld)* in the outage results. The *two-step approach* (B1) represents conventional map matching: the GNSS/IMU solution is computed first, and the reported position is then replaced by the closest point on the map, without feeding the correction back into the filter. The proposed one-step modes (C1–C3) use the same filter and association but apply the map as an observation that updates state and covariance (Table 6.1).

Table 6.1 Estimation modes. In the nominal experiment the one-step modes use the OSM map ($\sigma_{\mathrm{lat}} = 0.35\,\mathrm{m}$); in the outage experiment they are run with the OSM map and with the orthophoto map ($\sigma_{\mathrm{lat}} = 0.20\,\mathrm{m}$).

ID	Mode	Map update	Purpose
B0	GNSS/IMU	None	Primary sensor-fusion baseline
B1	Two-step	Output only	Separates snapping from filter feedback
C1	Map heading	Yaw	Tests orientation information
C2	Line normal	Cross-track	Tests coordinate information normal to the route
C3	Combined	Cross-track and yaw	Tests combined coordinate and orientation information

Every mode is run from a fresh initial state with identical sensor inputs and evaluation timestamps. B1 uses the same association as the one-step methods but changes only the reported position; C1–C3 update the filter state and covariance. In the nominal experiment, all map modes replay the association schedule of the GNSS/IMU baseline. In the outage experiment, each filter associates from its own state, because the baseline drifts by tens of metres and its schedule would not describe the constrained filters.

The outage windows were fixed before any outage result was inspected. They are 60 s long, start every 120 s from 100 s, and lie entirely inside the reference coverage (0 s to 676 s). Windows 1 to 4 each include a station stop.

7 Results

Two experiments are reported (Chapter 6). In the *nominal* experiment all GNSS observations are used and the OSM constraints are evaluated against the digitized evaluation reference. In the *outage* experiment GNSS is withheld in five 60 s outage windows, 1: 100 s to 160 s, 2: 220 s to 280 s, 3: 340 s to 400 s, 4: 460 s to 520 s, and 5: 580 s to 640 s after the start of the recording. The OSM and orthophoto constraints are compared there, and all errors are measured against the withheld GNSS positions. Windows 1 to 4 each contain a station stop, and the train runs at line speed through window 5.

An *epoch* is one GNSS measurement time; the recording has 780 epochs at 1 s spacing. The evaluation reference ends about 100 s before the end of the recording, so nominal accuracy values are computed over the 677 epochs it covers, the same epochs for every method. The outage values use the 299 epochs inside the five windows. In the nominal experiment the filter is evaluated at its track point, 3 m below the roof antenna (Equation (4.23)); in the outage experiment its antenna position is compared with the withheld GNSS. All numbers are taken from the runs listed in Table A.2. Each of the following sections first reports the nominal experiment and then the outages.

7.1 Trajectory Comparison

This section shows where the estimated trajectories lie relative to the track, as maps and as error profiles along the route. The heading angle itself is evaluated separately in Section 7.3.

7.1.1 Continuous GNSS

Under continuous GNSS, all trajectories lie within a few decimetres of the reference and cannot be told apart on a map. Figure 7.1 therefore shows the distance of the GNSS/IMU baseline to the reference along the whole route. Most of the time it stays below 0.1 m; the largest values, up to about 0.22 m, occur in curves and in the first 100 s. The five marked sections, labelled in time order, are the four 20-second sections in which the recorded GNSS lies farthest from the reference (B to E) and the curve with the largest map correction (A). In B to E the recorded GNSS is 0.21 m to 0.31 m from the reference, whereas the track point of the filter is only 0.03 m to 0.13 m from it. The difference is the antenna height: the GNSS measures the roof antenna, which moves sideways when the vehicle rolls on canted track (Section 4.3). The OSM constraints move the estimate further towards the reference by a few centimetres.

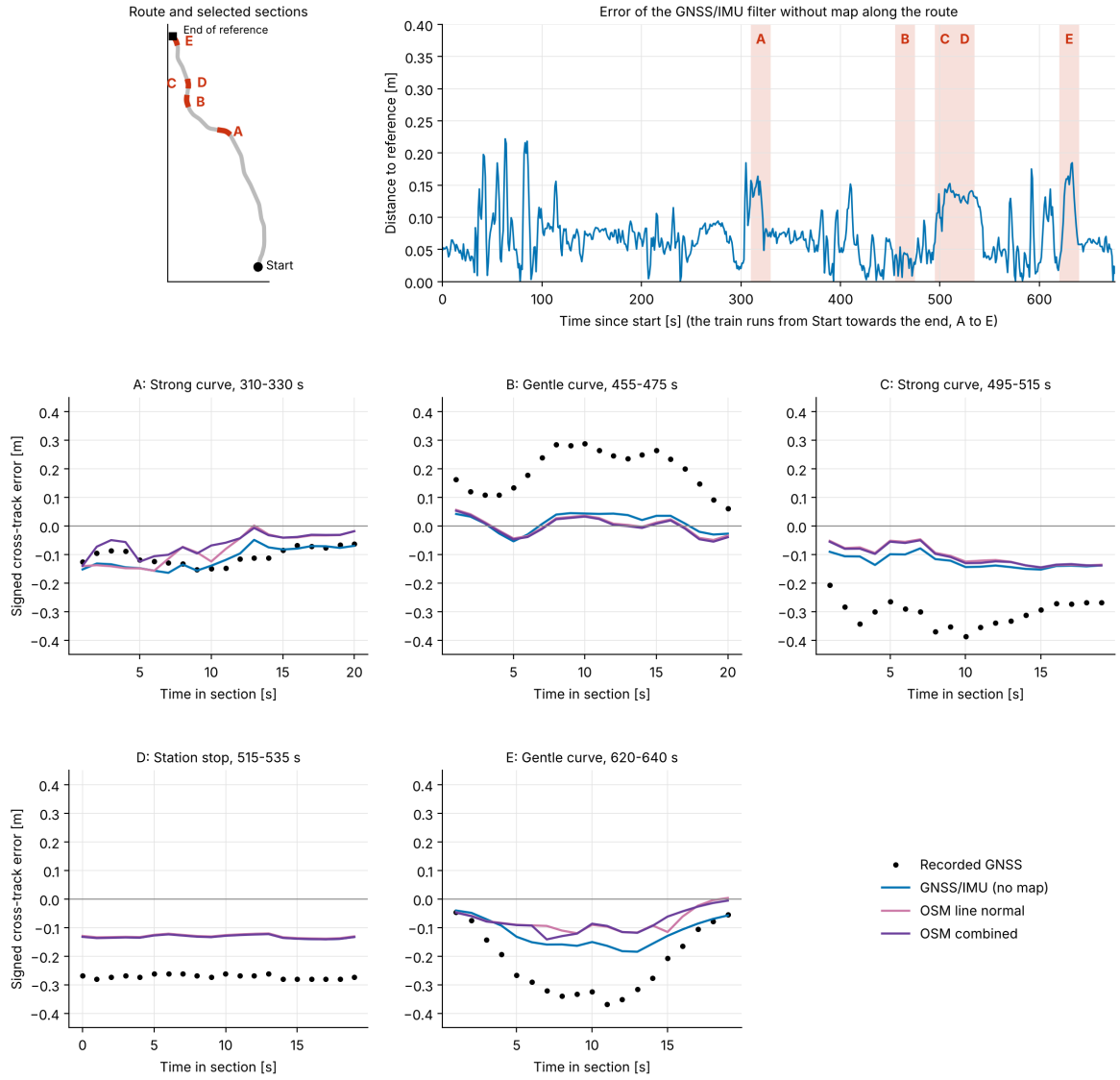


Figure 7.1 Nominal experiment. Top: GNSS/IMU baseline error against the digitized evaluation reference along the route, with the selected sections A to E, numbered in the order in which the train passes them from the start in the south. Bottom: signed cross-track error of the recorded GNSS (antenna) and of the baseline and the OSM line-normal and combined constraints (track point) in these sections. The line-normal and combined curves largely overlap.

7.1.2 GNSS Outages

The trajectories differ clearly during the outages (Figure 7.2). Without map updates, and with the heading observation alone, the estimate leaves the track by tens of metres within one window. With the line-normal constraint, the estimate stays on the track in every window, with the OSM map and with the orthophoto map. The OSM combined filter leaves the track in window 3. The constrained trajectories end short of or beyond the last withheld GNSS position: the position along the track is not corrected by the map.

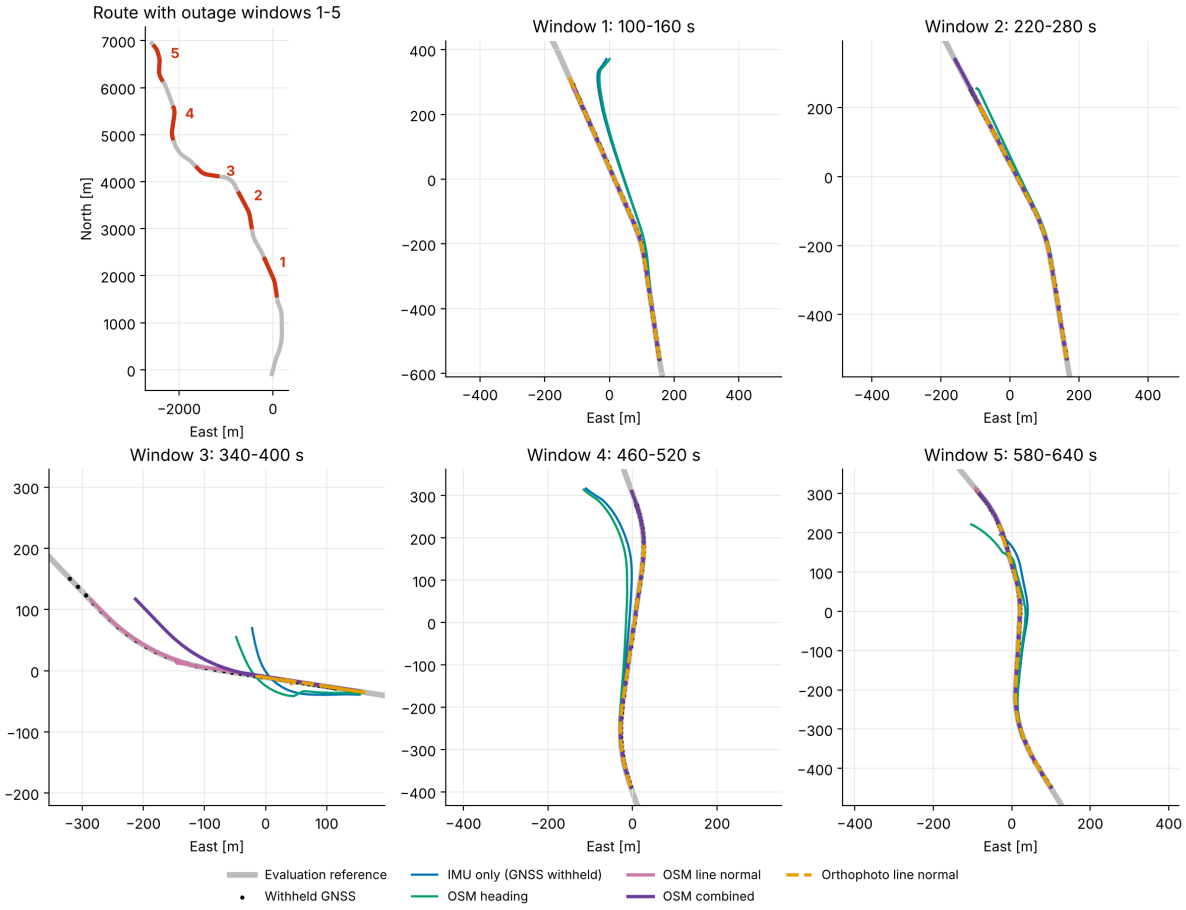


Figure 7.2 Outage experiment: estimated trajectories in the five outage windows, with the evaluation reference and the withheld GNSS positions. Top left: location of the windows along the route.

7.2 Position Accuracy

7.2.1 Continuous GNSS

Table 7.1 reports the perpendicular distance to the digitized evaluation reference in the nominal experiment. The mean describes the typical error, whereas the RMSE gives more weight to large errors. The 95% interval of the RMSE is obtained with a moving-block bootstrap. Figure 7.3 shows the same RMSE values with their intervals and how often each method stays within 5 cm, 10 cm and 15 cm of the reference.

Table 7.1 Nominal experiment, OSM map ($\sigma_{\mathrm{lat}} = 0.35\,\mathrm{m}$): perpendicular distance to the digitized evaluation reference over 677 epochs.

Method	Mean	Median	RMSE (m)	95th perc.	RMSE 95% CI (m)
Recorded GNSS positions	0.0794	0.0419	0.1182	0.2795	[0.0884, 0.1465]
GNSS/IMU ESKF	0.0710	0.0655	0.0816	0.1485	[0.0726, 0.0909]
Two-step output projection	0.3539	0.2348	0.4689	0.9697	[0.3676, 0.5556]
Map-heading observation	0.0716	0.0658	0.0821	0.1492	[0.0729, 0.0916]
Line-normal constraint	0.0585	0.0500	0.0731	0.1381	[0.0633, 0.0832]
Combined lateral position and heading	0.0546	0.0451	0.0709	0.1384	[0.0597, 0.0818]

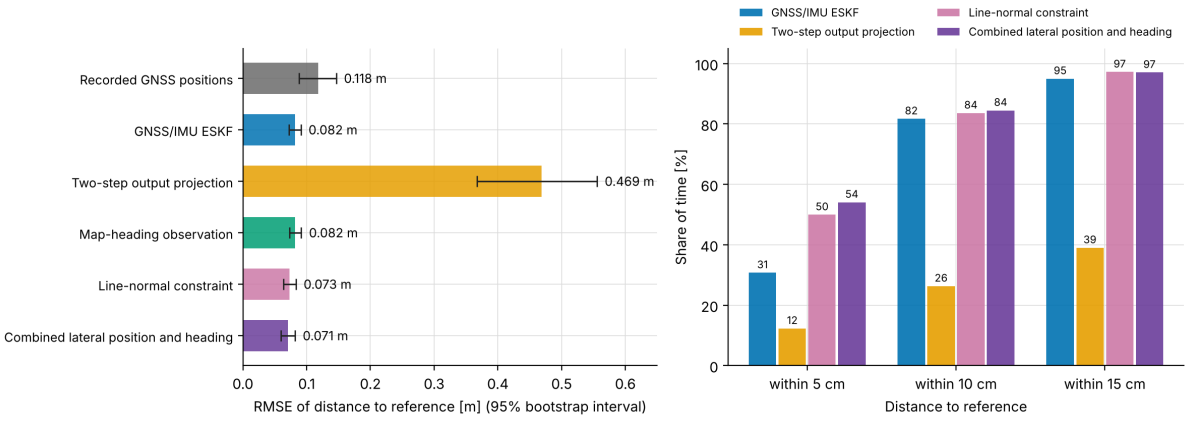


Figure 7.3 Nominal experiment, OSM map. Left: RMSE of the distance to the digitized evaluation reference with 95% bootstrap interval. Right: share of the 677 epochs at which the estimate lies within 5 cm, 10 cm and 15 cm of the reference. The map-heading observation is not drawn because it coincides with the GNSS/IMU filter.

The GNSS/IMU baseline reaches 0.082 m, against 0.118 m for the recorded RTK GNSS, because it reports the track point and not the roof antenna; its 95th percentile is about half that of the GNSS. The line-normal constraint reduces the RMSE further to 0.073 m (10.4 %) and the combined observation to 0.071 m (13.1 %). The confidence intervals still overlap, but the position constraints shift the whole error distribution towards smaller values: the combined filter is within 5 cm of the reference at 54 % of the epochs, against 31 % without map (Figure 7.3). The two-step projection reaches 0.469 m: it replaces the reported position by the closest OSM point, so its error contains the OSM error itself. The one-step constraint weights the map through its observation covariance and does not force the state onto the line.

The OSM map provides an association at 777 of the 780 epochs; three are rejected as ambiguous. The innovation gate accepts 773 of the remaining heading observations, 720 line-normal observations, and 760 combined observations.

7.2.2 GNSS Outages

Table 7.2 summarizes the outage experiment over all 299 outage epochs, and Figure 7.4 breaks the cross-track and along-track errors down by window. Without map updates, the cross-track error reaches 35 m RMSE and the horizontal error up to 309 m. The line-normal constraint keeps the cross-track error at 0.55 m RMSE with OSM and 0.57 m with the orthophoto map, a reduction of about 98 % with both maps. With OSM it also halves the horizontal error; with the orthophoto map the horizontal error stays at the level of the baseline, because it is dominated by the along-track error (Section 7.4.2).

Table 7.2 Outage experiment: errors against the withheld GNSS over all 299 outage epochs. Inside the windows the filter without map runs on the IMU alone. Change: cross-track RMSE relative to IMU only.

Map	Method	Horizontal RMSE (m)	Along RMSE (m)	Cross RMSE (m)	Change cross	Heading RMSE (deg)	Accepted updates
None	IMU only	76.95	70.64	35.30	–	2.37	0
OSM	Map heading	73.93	64.91	38.89	+10%	1.33	211
OSM	Line normal	38.64	38.64	0.55	-98%	1.82	299
OSM	Combined	46.13	45.93	7.30	-79%	1.06	284
Orthophoto	Map heading	74.39	65.52	40.80	+16%	1.79	197
Orthophoto	Line normal	81.50	82.53	0.57	-98%	4.41	272
Orthophoto	Combined	65.80	67.75	10.36	-71%	2.14	239

The per-window view shows two separate effects. The cross-track error of both line-normal filters stays below 1 m in all five windows, one to two orders of magnitude below IMU only (95 % to 99.8 % lower). The combined filters do the same in most windows but lose the map in window 3, and with the orthophoto map their error also grows to 1 m and 3 m in windows 4 and 5. The along-track error, in contrast, remains tens of metres for every method. It does not follow the cross-track error and is discussed in Section 7.4.2.

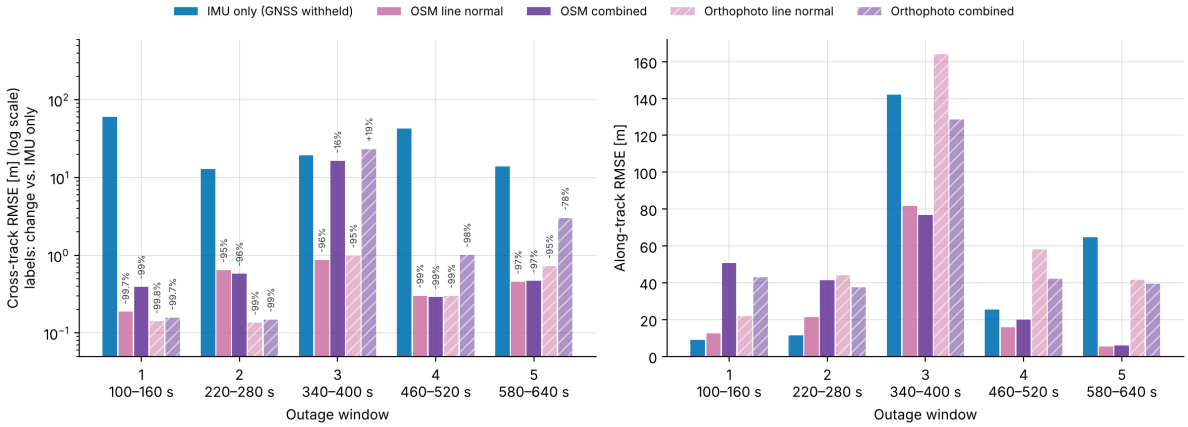


Figure 7.4 Outage experiment: cross-track RMSE (left, logarithmic axis) and along-track RMSE (right) against the withheld GNSS in each outage window. Hatched bars: orthophoto map. The labels give the change of the cross-track RMSE relative to IMU only.

7.3 Orientation Accuracy

7.3.1 Continuous GNSS

Table 7.3 reports the heading error of the nominal experiment, defined in Equation (6.2), over the 554 of the 677 epochs with a recorded speed of at least 2 m/s.

Table 7.3 Nominal experiment, OSM map: heading error against the direction of the digitized evaluation reference, 554 epochs.

Method	Mean abs.	RMSE	95th perc. (deg)	Signed mean
GNSS/IMU ESKF	1.23	1.89	3.62	0.79
Map-heading observation	0.33	0.49	1.02	0.00
Line-normal constraint	1.28	1.92	3.73	0.91
Combined	0.34	0.50	1.00	−0.01

The GNSS/IMU baseline has a heading RMSE of 1.89° and a mean bias of 0.8°. The largest errors, up to 11°, occur in the first minute, while the train accelerates after the start and the yaw estimate is still converging (Figure 7.5); GNSS positions determine the heading only indirectly. With the heading observation the RMSE falls to 0.49° (74 % lower) and the bias disappears; the combined observation reaches almost the same value (0.50°). The line-normal constraint leaves the heading unchanged.

7.3.2 GNSS Outages

During the outages (Table 7.2), the heading observation reduces the heading RMSE from 2.37° without map to 1.33° with OSM (44 % lower) and 1.79° with the orthophoto map; the OSM combined filter reaches 1.06°. Figure 7.6 shows the error inside each window. Without map, the heading drifts with the gyroscope bias, most clearly in window 4, where it reaches about –5°.

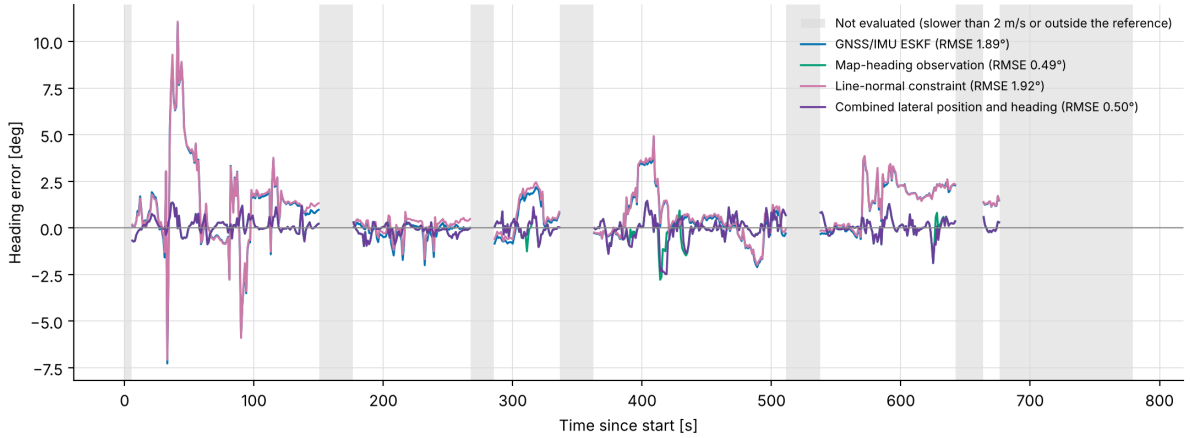


Figure 7.5 Heading error over time, nominal experiment, OSM map. The line-normal curve overlaps the GNSS/IMU curve and the combined curve largely overlaps the map-heading curve. Grey: epochs that are not evaluated because the train is slower than 2 m/s (station stops) or outside the reference (after about 677 s).

With either map, the heading error stays within about 2° in windows 1, 2 and 4, and the two map curves are close to each other.

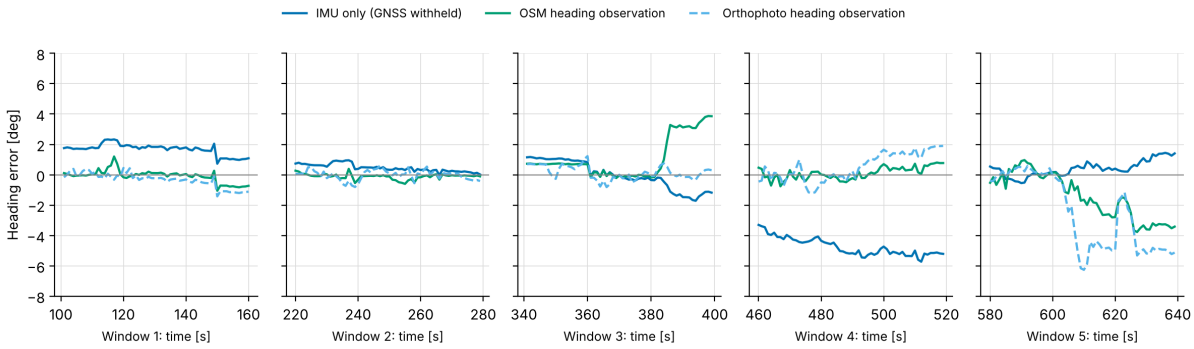


Figure 7.6 Outage experiment: heading error against the direction of the withheld GNSS path in the five outage windows, without map and with the map-heading observation from OSM and from the orthophoto map.

Windows 3 and 5 show the limit of the heading observation. The map direction is taken at the estimated position, not at the true one. In window 3 the heading-only filters fall more than 130 m behind the train after the station stop. The OSM filter then takes the direction of a straight section while the train is already in a curve, and its heading error grows to about 4° at the end of the window. In window 5 the heading-only filters drift by 20 m to 35 m sideways and 38 m to 46 m along the track (Figure 7.4). On this section, which contains several curves, the map again supplies the direction of a different part of the track, and the heading error grows to 3° to 6° with both maps, more than without map. A heading observation is only as good as the position at which it is looked up.

7.4 Effect of Position Constraints

7.4.1 Continuous GNSS

Under continuous GNSS, the line-normal constraint acts on errors of a few centimetres to about 0.15 m. Table 7.4 lists the sections of Figure 7.1.

Table 7.4 Nominal experiment, OSM map: cross-track RMSE (m) against the digitized evaluation reference in the sections A to E of Figure 7.1.

	Interval (s)	Geometry	Rec. GNSS	GNSS/IMU	Heading	Line n.	Comb.
A	310–330	Strong curve	0.109	0.118	0.117	0.096	0.070
B	455–475	Gentle curve	0.207	0.034	0.033	0.030	0.031
C	495–515	Strong curve	0.309	0.127	0.129	0.109	0.111
D	515–535	Station stop	0.272	0.132	0.135	0.131	0.133
E	620–640	Gentle curve	0.249	0.130	0.132	0.087	0.088

In sections B to E most of the GNSS offset is removed by the antenna height alone: the baseline lies 0.03 m to 0.13 m from the reference, against 0.21 m to 0.31 m for the recorded GNSS. The map constraints reduce the remaining error further in the curves. In C the line-normal constraint lowers it from 0.127 m to 0.109 m (14 %), in E from 0.130 m to 0.087 m (33 %), and in A the combined observation lowers it from 0.118 m to 0.070 m (41 %). In B the baseline is already within 4 cm of the reference. At the station stop D the constraints change nothing; the remaining offset of 0.13 m is constant while the train stands and is most likely a local difference between the digitized reference and the track.

Figure 7.7 and Table 7.5 extend this view to the whole route. Every evaluated epoch is assigned to a straight section, a curve with a radius below 1500 m, or a standstill below 1 m/s.

Table 7.5 Nominal experiment, OSM map: cross-track RMSE against the digitized evaluation reference (cm) and heading RMSE (deg) by track geometry. Heading is evaluated above 2 m/s only.

	Epochs	Rec. GNSS	GNSS/IMU	Heading	Line n.	Comb.
Cross-track RMSE (cm)						
Straight	338	4.1	6.2	6.3	5.3	5.0
Curve	228	17.4	10.0	10.1	9.2	9.2
Standstill	111	13.4	9.0	9.1	8.0	7.4
Heading RMSE (deg)						
Straight	327		1.57	0.45	1.60	0.47
Curve	227		2.27	0.54	2.31	0.55

In curves, the antenna height reduces the cross-track RMSE from 17.4 cm for the recorded GNSS to 10.0 cm for the baseline, and the position constraints reduce it further to 9.2 cm. On straight track the baseline (6.2 cm) is slightly worse than the recorded GNSS (4.1 cm). There the antenna is directly above the track, and a small error of the estimated roll moves the computed track point sideways (0.5 ° correspond to 2.6 cm at $h = 3\,\mathrm{m}$); the constraints reduce this to 5.0 cm to 5.3 cm. At standstill they reduce the error from 9.0 cm to 7.4 cm to 8.0 cm. The map makes the solution worse only where it is less accurate than the GNSS: between about 400 s and 440 s, the OSM line lies about 0.7 m from the reference while the baseline is within 0.08 m, and the constraints move the estimate 2 cm to 3 cm away from the reference (red epochs in Figure 7.7).

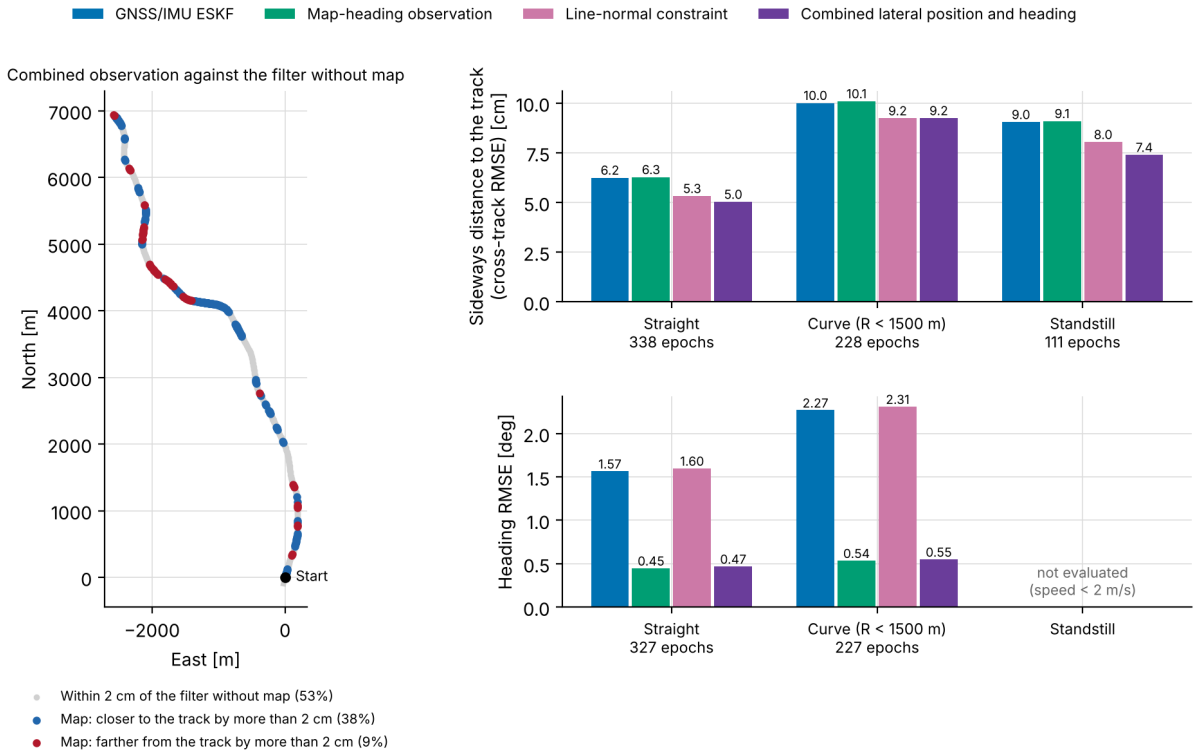


Figure 7.7 Nominal experiment, OSM map. Left: epochs at which the combined observation brings the estimate more than 2 cm closer to the reference than the filter without map (blue), more than 2 cm farther away (red), or changes it by less (grey). Right: cross-track and heading RMSE by track geometry.

The heading observation helps on straight track and in curves alike, because the heading of the GNSS/IMU solution is poorly determined everywhere compared with the direction of the map.

7.4.2 GNSS Outages and Map Lock

The position constraint is decisive during the outages. Every map observation first passes an *innovation gate* (Section 4.7): its innovation is compared with the uncertainty the filter expects for it, and an observation that is too large to be plausible is rejected as an outlier. A filter is said to keep *map lock* when its map observations keep passing this gate during the window. It *loses map lock* when the estimate has drifted so far that every map observation is rejected; from then on the map is not used and the error grows as without map. Table 7.6 lists the cross-track error per window for both maps, and Figures 7.8 and 7.10 show the error over time.

Table 7.6 Outage experiment: cross-track RMSE (m) against the withheld GNSS per outage window. Orthophoto availability is the share of window epochs at which the orthophoto map has a digitized segment.

Window	1	2	3	4	5
Interval (s)	100–160	220–280	340–400	460–520	580–640
IMU only (GNSS withheld)	60.41	12.96	19.43	42.79	13.83
OSM line normal	0.19	0.64	0.87	0.30	0.46
OSM combined	0.39	0.58	16.42	0.29	0.47
Orthophoto line normal	0.14	0.13	0.99	0.30	0.72
Orthophoto combined	0.16	0.15	23.10	1.01	3.00
Orthophoto availability	98%	100%	100%	70%	82%

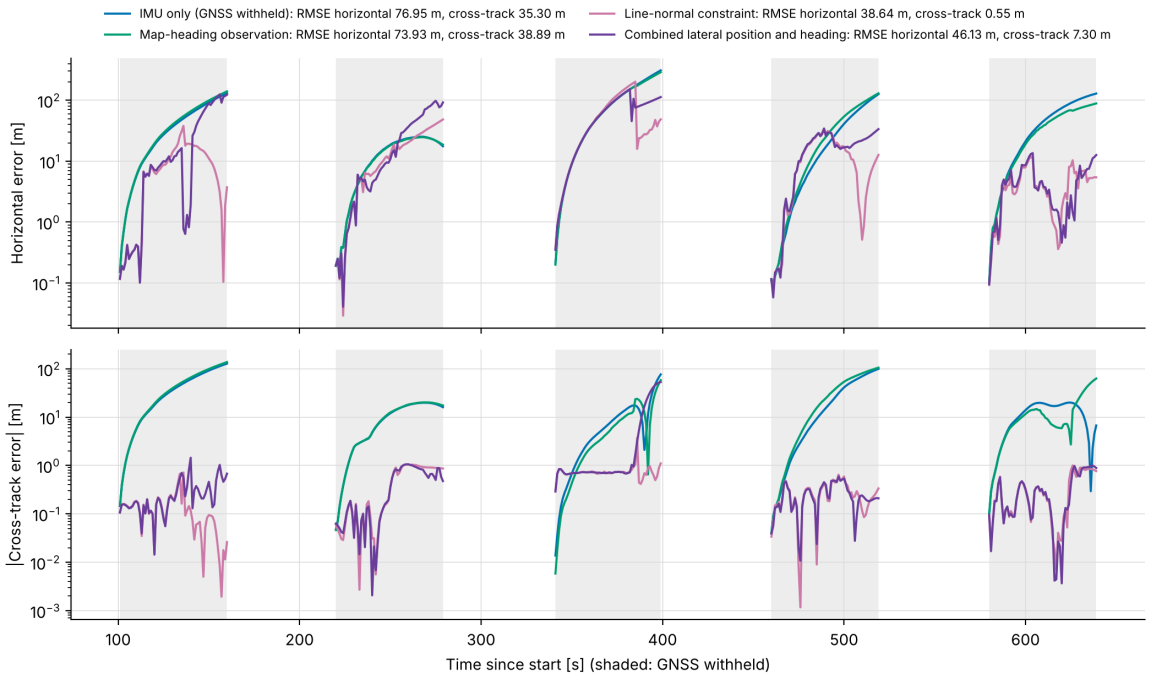


Figure 7.8 Outage experiment, OSM map: horizontal and cross-track error against the withheld GNSS. Shaded: outage windows. Logarithmic axes.

Both line-normal filters keep map lock in all five windows. With OSM, the cross-track error stays between 0.19 m and 0.87 m. With the orthophoto map it is lower in the first two windows, 0.13 m to 0.14 m, which reflects the higher accuracy of the orthophoto digitization, and 0.30 m to 0.99 m in windows 3 to 5. Windows 4 and 5 contain digitization gaps (70% and 82% availability). While the map is missing, the IMU drift continues, but it stays small enough for the gate when the map returns. The combined filters are less robust: both lose map lock in window 3, where the train departs from a station into a curve (Section 7.5.2).

Figure 7.9 places these errors on the track. Windows 1 and 2 are mostly straight, windows 4 and 5 lie between 58 % and 72 % in curves, and window 3 starts with a 20 s station stop followed by the sharpest curve of the five windows. The line-normal filters hold the cross-track error between about 0.01 m and 1 m on straight track and in curves alike, so the curves themselves are not the problem. The larger errors appear at two kinds of places: in the sharp curve of window 3, where the combined filters lose map lock, and directly after gaps of the orthophoto map in windows 4 and 5, where the orthophoto combined filter takes a heading error of up to 14 °.

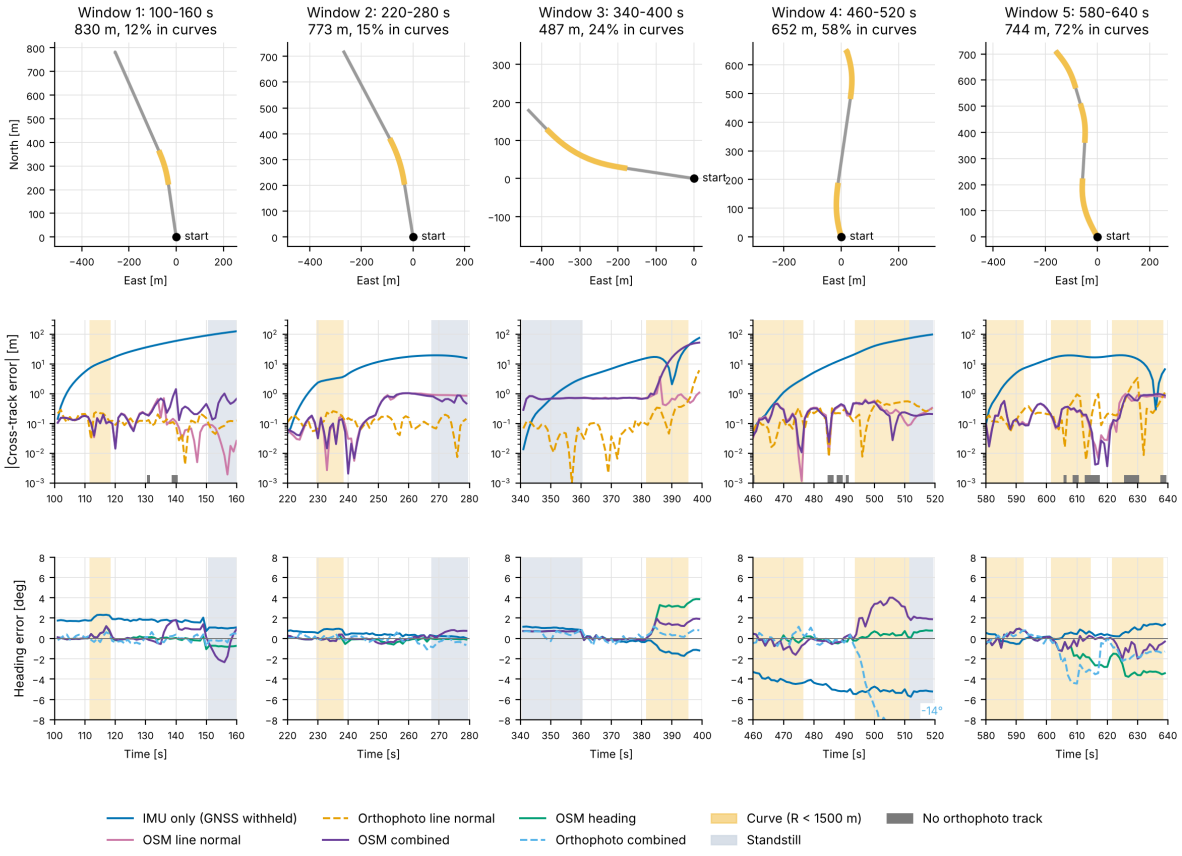


Figure 7.9 Outage experiment: track shape (top), cross-track error (middle, logarithmic) and heading error (bottom) in each window, with curves (radius below 1500 m), standstill and epochs without orthophoto track marked. Heading errors beyond 8 ° are labelled with their extreme value.

No position constraint observes the along-track position: a track line tells the filter how far it is from the track, but not where along the track it is. The along-track error is therefore not controlled (Figure 7.4, right). With the OSM line-normal constraint it is larger than without map in windows 1 and 2 (13 m and 21 m against 9 m and 11 m) and smaller in windows 3 to 5, because the lateral corrections also change the velocity through the state covariance. Over all outage epochs it falls from 71 m to 39 m, but it remains tens of metres; with the orthophoto line-normal filter it

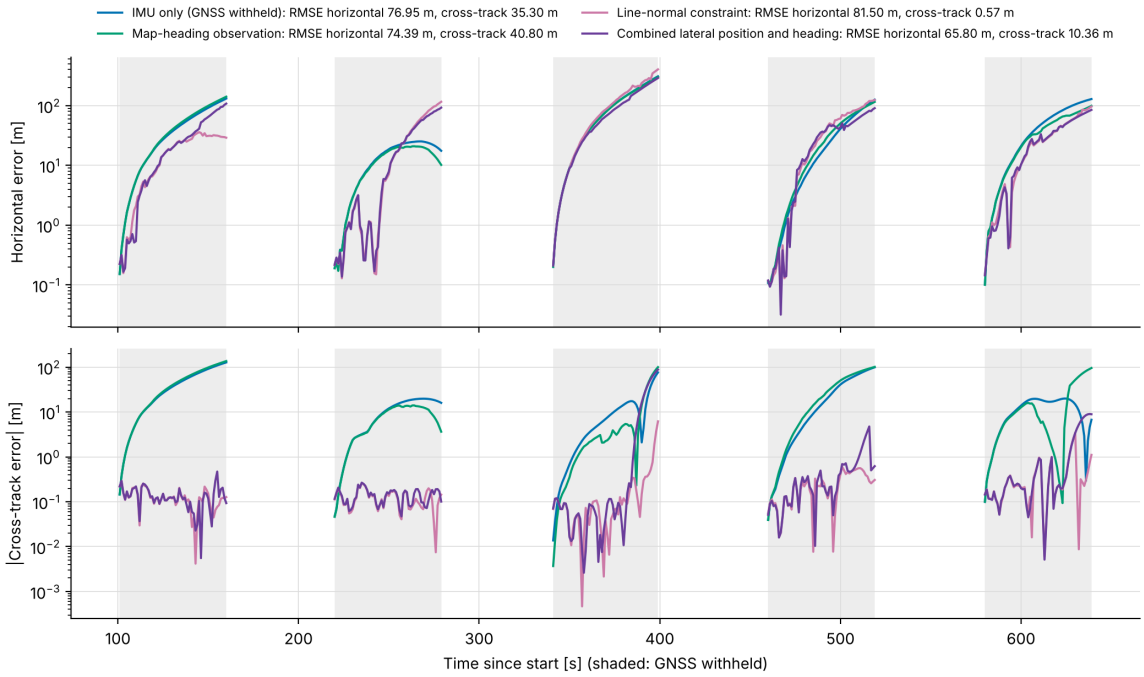


Figure 7.10 Outage experiment, orthophoto map: horizontal and cross-track error against the withheld GNSS. Shaded: outage windows. Logarithmic axes.

is 83 m. In window 3, where the train departs from a station, it exceeds 100 m for every filter except the OSM line-normal and combined filters (82 m and 77 m).

7.5 Effect of Orientation Constraints

7.5.1 Continuous GNSS

The heading observation and the line-normal constraint act on different states. Under continuous GNSS, the heading observation reduces the heading RMSE by about three quarters (Table 7.3) but changes the position RMSE by less than 1 mm (Table 7.1). On a straight segment, a correct heading does not tell the filter where the track is laterally; only the line-normal residual supplies that coordinate. Conversely, the line-normal constraint does not improve the heading. The combined observation obtains both effects; in curve A it also gives the lowest position error (Table 7.4).

7.5.2 GNSS Outages

During the outages, the heading observation alone does not limit the position drift (Table 7.2): the position offset that builds up from velocity and acceleration errors is not removed by a correct heading. In combination with the line-normal constraint, the heading residual makes the update stricter. In window 3, both combined filters lose map lock (16.4 m with OSM and 23.1 m with the orthophoto map, against 0.87 m and 0.99 m for the line-normal filters). The reason is the same as for the heading-only filter in Section 7.3.2: with a large along-track error the map direction is

looked up on the wrong part of the track, and the stacked observation fails the two-dimensional gate. With OSM the combined filter accepts 284 instead of 299 updates. With the orthophoto map it also has larger errors than the line-normal filter in windows 4 and 5 (1.0 m and 3.0 m against 0.30 m and 0.72 m).

7.6 Discussion

Under continuous RTK GNSS, the largest single improvement does not come from the map but from describing the antenna correctly. Reporting the track point 3 m below the roof antenna reduces the RMSE against the reference from 0.118 m to 0.082 m and removes most of the decimetre offsets in curves. On this basis the OSM line-normal constraint improves the position by a further 10 % over the route and by up to 33 % in the curves with the largest errors, and the heading observation reduces the heading error by three quarters. The constraints help where the GNSS/IMU solution is far from the track and slightly degrade it where the map is less accurate than the GNSS. The two-step projection shows that snapping an accurate trajectory to an inaccurate map makes it worse, which the one-step approach avoids because the map is weighted against the other sensors.

The value of the one-step approach appears when GNSS is unavailable. The line-normal constraint then reduces the cross-track drift from tens of metres to about half a metre with both maps (about 98 % lower), and to 0.13 m to 0.14 m with the orthophoto map where it is dense; an output-only correction cannot do this because it never corrects the filter state. Two parts of the filter are needed for this robustness: the track point, so that the map refers to the track and not to the antenna, and the covariance reset on GNSS reacquisition, so that the large correction after an outage does not corrupt the attitude before the next window. Three limitations remain. First, a track line gives no along-track information, so the along-track error is not controlled; along-track observations from discrete map features such as switches, platform ends, or curvature changes would be needed. Second, along-track error becomes a lateral and directional inconsistency in curves; the combined observations then fail the gate and lose the map, so the line-normal observation alone is the more robust choice during outages. Third, the antenna height is an approximate value, and errors of the estimated roll move the computed track point sideways by a few centimetres.

All results come from one railway recording. No claim is made about other recordings or outages longer than 60 s.

8 Conclusion and Future Work

8.1 Summary

This thesis implemented a one-step integration of map data into a GNSS/IMU ESKF, with the map treated as an additional sensor. Heading, line-normal, and combined observations were derived from OSM ways and from a track polyline digitized from true orthophotos. The map observations and the evaluation refer to the track point 3 m below the roof antenna. For the orthophoto map, the digitization was shifted laterally by half the track gauge to the track centre line, split at digitization gaps, and associated only through perpendicular residuals. All configurations were compared with the GNSS/IMU baseline and a two-step output projection. Under continuous GNSS, the OSM constraints were evaluated against a digitized evaluation reference; during predefined GNSS outages, the OSM and orthophoto constraints were compared against the withheld GNSS positions.

8.2 Main Findings

1. **Antenna height.** The GNSS antenna is on the roof, about 3 m above the track, and moves sideways when the vehicle rolls on canted track. Reporting the track point below the antenna reduces the RMSE of the GNSS/IMU solution against the reference from 0.118 m to 0.082 m and removes most of the decimetre offsets in curves.
2. **Continuous GNSS.** On this basis, the OSM line-normal observation reduces the RMSE to 0.073 m (10 %) and the combined observation to 0.071 m (13 %). In the curves with the largest errors the gain is up to 33 % with the line-normal and 41 % with the combined observation. The bootstrap intervals overlap, so the route-wide gain is modest. The constraints help where the GNSS/IMU solution is far from the track, mainly in curves, and move the estimate a few centimetres away from the reference where OSM is less accurate than the GNSS.
3. **Orientation.** The heading observation does not change position but reduces the heading RMSE from 1.89 ° to 0.49 ° (74 %) and removes the baseline bias. During outages it is only as good as the position at which the map direction is looked up. Heading and line-normal information are complementary; the combined observation obtains both under continuous GNSS.
4. **One step versus two steps.** Output projection onto OSM raises the error to 0.469 m, because the reported position takes over the OSM error. Two-step correction is only as good as the map, and it cannot correct the filter state during outages.
5. **GNSS outages.** The line-normal observation keeps the cross-track error at 0.55 m RMSE with OSM and 0.57 m with the orthophoto map, compared with 35 m with the IMU alone, a reduction of about 98 %. It stays below 1 m and keeps map lock in all five windows with both maps, on straight track and in curves. This requires the covariance reset on GNSS reacquisition; without it, the correction after one outage corrupted the attitude before the next. The along-track error is not observed by a track line and stays large.

6. **Map accuracy.** Where it is dense, the orthophoto map gives a lower cross-track error than OSM (0.13 m to 0.14 m against 0.19 m to 0.64 m in the first two windows), and the line-normal filter bridges its digitization gaps. The combined observations are less robust than the line-normal observation alone: with both maps they lose map lock in window 3.

8.3 Limitations

All results come from one railway recording with nearly continuous RTK GNSS; outages were simulated. The nominal evaluation relies on a digitized reference whose own accuracy is limited by the 0.20 m orthophoto resolution. The antenna height of 3 m is an approximate value, and errors of the estimated roll move the computed track point sideways by a few centimetres, which is visible on straight track. The filter uses flat-Earth dynamics in UTM and engineering process-noise values. Only an ESKF was implemented.

8.4 Future Research Directions

- Along-track observations from discrete map features, such as switches, platform ends, and curvature changes, to bound the along-track drift.
- Re-association and gate inflation after repeated rejections, so that a filter can recover the map after a gap or a large drift.
- Filling orthophoto digitization gaps with OSM, combining accuracy and continuity.
- Additional recordings of the campaign and real GNSS degradation, such as forest sections and cuttings.
- The track cant from the surveyed alignment of the line, to fix the roll and thus the offset between antenna and track, and a correlated map-error model that would allow the antenna offset to be modelled as a lever arm inside the GNSS update.
- Repetition with an unscented Kalman filter and with an independently surveyed reference.

Bibliography

Atia, M. M. and Waslander, S. L. (2019). “Map-Aided Adaptive GNSS/IMU Sensor Fusion Scheme for Robust Urban Navigation”. In: *Measurement* 131, pp. 615–627. URL: https://www.sciencedirect.com/science/article/pii/S0263224118307899.

Bar-Shalom, Y., Li, X. R., and Kirubarajan, T. (2001). *Estimation with Applications to Tracking and Navigation*. Wiley. ISBN: 9780471416555. DOI: 10.1002/0471221279.

Brown, R. G. and Hwang, P. Y. C. (2012). *Introduction to Random Signals and Applied Kalman Filtering*. 4th ed. Wiley. ISBN: 9780470609699.

El Najjar, M. E. and Bonnifait, P. (2005). “A Road-Matching Method for Precise Vehicle Localization Using Belief Theory and Kalman Filtering”. In: *Autonomous Robots* 19.2, pp. 173–191. DOI: 10.1007/s10514-005-0609-1.

Falco, G., Pini, M., and Marucco, G. (2017). “Loose and Tight GNSS/INS Integrations: Comparison of Performance Assessed in Real Urban Scenarios”. In: *Sensors* 17.2, p. 255. DOI: 10.3390/s17020255.

Fervers, F., Bullinger, S., Bodensteiner, C., Arens, M., and Stiefelhagen, R. (2022). *Continuous Self-Localization on Aerial Images Using Visual and LiDAR Sensors*. arXiv preprint. DOI: 10.48550/arXiv.2203.03334.

– (2023). “Uncertainty-Aware Vision-Based Metric Cross-View Geolocalization”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Vancouver, pp. 21621–21631. URL: https://openaccess.thecvf.com/content/CVPR2023/html/Fervers_Uncertainty-Aware_Vision-Based_Metric_Cross-View_Geolocalization_CVPR_2023_paper.html.

Fouque, C., Bonnifait, P., and Bétaille, D. (2008). “Enhancement of Global Vehicle Localization Using Navigable Road Maps and Dead-Reckoning”. In: *IEEE/ION Position, Location and Navigation Symposium (PLANS)*. Monterey, CA. URL: https://hal.science/hal-00445289.

Frolow, R., Zhang, L., and Schwieger, V. (2026). “Precise Train Positioning With Unscented Kalman Filter and Low-Cost Sensors”. In: *IEEE Open Journal of Intelligent Transportation Systems* 7, pp. 304–312. DOI: 10.1109/OJITS.2026.3652748.

Groves, P. D. (2013). *Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems*. 2nd ed. Boston: Artech House. ISBN: 9781608070053.

Gupta, N. and Hauser, R. (2007). *Kalman Filtering with Equality and Inequality State Constraints*. arXiv preprint. DOI: 10.48550/arXiv.0709.2791.

Haklay, M. and Weber, P. (2008). “OpenStreetMap: User-Generated Street Maps”. In: *IEEE Pervasive Computing* 7.4, pp. 12–18. DOI: 10.1109/MPRV.2008.80.

Heirich, O. (2016). “Bayesian Train Localization with Particle Filter, Loosely Coupled GNSS, IMU, and a Track Map”. In: *Journal of Sensors* 2016, p. 2672640. DOI: 10.1155/2016/2672640.

Julier, S. J. and LaViola, J. J. (2007). “On Kalman Filtering with Nonlinear Equality Constraints”. In: *IEEE Transactions on Signal Processing* 55.6, pp. 2774–2784. DOI: 10.1109/TSP.2007.893949.

Kalman, R. E. (1960). “A New Approach to Linear Filtering and Prediction Problems”. In: *Journal of Basic Engineering* 82.1, pp. 35–45. DOI: 10.1115/1.3662552.

Landesamt fuer Geoinformation und Landentwicklung Baden-Wuerttemberg (2026). *Digital Orthophotos with 20 cm Ground Resolution*. URL: https://www.lgl-bw.de/Produkte/Geodatendienste/Luftbildprodukte/index.html (visited on 11/09/2026).

Miller, M. M., Blasch, E., Nguyen, T. Q., and Yang, C. (2007). “Constrained Estimation for GPS/Digital Map Integration”. In: *Proceedings of the 2007 National Technical Meeting of The Institute of Navigation*. San Diego, CA, pp. 1119–1127. URL: https://www.ion.org/publications/abstract.cfm?articleID=7207.

Newson, P. and Krumm, J. (2009). “Hidden Markov Map Matching Through Noise and Sparseness”. In: *Proceedings of the 17th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, pp. 336–343. DOI: 10.1145/1653771.1653818.

Quddus, M. A., Ochieng, W. Y., and Noland, R. B. (2007). “Current Map-Matching Algorithms for Transport Applications: State-of-the Art and Future Research Directions”. In: *Transportation Research Part C: Emerging Technologies* 15.5, pp. 312–328. DOI: 10.1016/j.trc.2007.05.002.

Quddus, M. A., Ochieng, W. Y., Zhao, L., and Noland, R. B. (2003). “A General Map Matching Algorithm for Transport Telematics Applications”. In: *GPS Solutions* 7.3, pp. 157–167. DOI: 10.1007/s10291-003-0069-z.

Simon, D. and Chia, T. L. (2002). “Kalman Filtering with State Equality Constraints”. In: *IEEE Transactions on Aerospace and Electronic Systems* 38.1, pp. 128–136. DOI: 10.1109/7.993234.

Solà, J. (2017). *Quaternion Kinematics for the Error-State Kalman Filter*. arXiv preprint. DOI: 10.48550/arXiv.1711.02508.

Studholme, C., Hill, D. L. G., and Hawkes, D. J. (1999). “An Overlap Invariant Entropy Measure of 3D Medical Image Alignment”. In: *Pattern Recognition* 32.1, pp. 71–86. DOI: 10.1016/S0031-3203(98)00091-0.

Tao, Z., Bonnifait, P., Frémont, V., Ibañez-Guzmán, J., and Bonnet, S. (2017). “Road-Centred Map-Aided Localization for Driverless Cars Using Single-Frequency GNSS Receivers”. In: *Journal of Field Robotics* 34.5, pp. 1010–1033. DOI: 10.1002/rob.21708.

Titterton, D. H. and Weston, J. L. (2004). *Strapdown Inertial Navigation Technology*. 2nd ed. Institution of Engineering and Technology. ISBN: 9780863413582. DOI: 10.1049/PBRA017E.

Veronese, L., Aguiar, E. de, Nascimento, R. C., Guivant, J., Auat Cheein, F. A., De Souza, A. F., and Oliveira-Santos, T. (2015). “Re-emission and Satellite Aerial Maps Applied to Vehicle Localization on Urban Environments”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Hamburg, pp. 4285–4290. DOI: 10.1109/IROS.2015.7353984.

Vora, A., Agarwal, S., Pandey, G., and McBride, J. (2020). *Aerial Imagery Based LiDAR Localization for Autonomous Vehicles*. arXiv preprint. DOI: 10.48550/arXiv.2003.11192.

Winter, H., Willert, V., and Adamy, J. (2019). *Train-Borne Localization Exploiting Track-Geometry Constraints – A Practical Evaluation*. arXiv preprint. DOI: 10.48550/arXiv.1906.07569.

A Annex

A.1 Implementation Parameter Register

Table A.1 Parameters of the final runs.

Parameter	Value	Note
GNSS position std. (mode 4 / 5 / other)	0.10 / 0.50 / 5 m	Adaptive model of Frolow et al. (2026)
Maximum association distance	30 m	Both maps
Association ambiguity margin	2	Ambiguous associations rejected
OSM lateral std.	0.35 m	Median GNSS-to-OSM distance
Orthophoto lateral std.	0.20 m	One orthophoto pixel
Orthophoto lateral offset	0.7175 m	Half the standard gauge
Orthophoto gap length	10 m	Chord error < 7 cm at $R = 188$ m
Orthophoto along-residual tolerance	1 m	Rejects gap associations
Map heading std. (both maps)	1 °	RMS segment-to-GNSS direction difference
Antenna height above the track h	3 m	Track point, Equation (4.23)
Covariance reset after an outage	On	Equation (4.24)
Minimum speed for heading metric	2 m/s	Evaluation only
NIS gate (1-D / 2-D)	6.6349 / 9.2103	99% chi-square
Reference coverage distance	20 m	677 of 780 epochs covered
Bootstrap block length	20 epochs	Moving block
Outage windows	60 s, 5 windows	Every 120 s from 100 s

A.2 Run Register

All runs were produced with `run_final.py`, which stores the exact command, the software revision, and the full log in each output directory under `outputs/final_runs/railway/`.

Table A.2 Runs used in Chapter 7.

Run directory	Map	Content
`20261109_060855_osm`	OSM	Nominal, fixed association schedule
`20261109_061039_osm_outage`	OSM	Five outages, end-to-end association
`20261109_061210_orthophoto_outage`	Orthophoto	Five outages, end-to-end association

A.3 Declaration on the Use of AI Tools or AI Systems

The following register follows the disclosure structure of the IIGS thesis template. The methods, implementation, experiments, and evaluation were designed and carried out by the author; AI tools were used for supporting tasks only.

Table A.3 AI tools used in preparing the thesis and implementation.

No.	Tool type, name, version, and year	Purpose and extent of use
1	Large language model: OpenAI ChatGPT 5.6, 2026	Improving sentence structure and wording of the text, writing and tidying code comments, support with plotting and visualization scripts, and support in literature search. All text, references, code, and results were checked by the author.
2	Coding agent / large language model: OpenAI Codex based on GPT-5, 2026	LaTeX template migration, document-formatting support, and assistance with implementation review and debugging. The author retains responsibility for the final code and results.